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Pre-test probability of stable coronary artery disease: a systematic review suggesting how to improve validation procedures

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Pre-test probability of stable coronary artery disease: a systematic review suggesting how to improve validation procedures

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ABSTRACT

Objective: Externally validated pretest probability models for the diagnosis of stable coronary artery disease determined through invasive coronary angiography or coronary computed tomography angiography, are addressed to characterize the best procedures in terms of predictive variables, discriminatory ability and methods completeness.

Design: Systematic review conforming to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.

Data sources: Global Health, Healthstar and MEDLINE® were systematically searched on 22 April 2020.

Eligibility criteria: Full papers published in English or Italian. Reasons of exclusion were (i) acute coronary syndrome, unstable chest pain, a history of myocardial infarction or previous revascularisation; (ii) models referring to diagnostic procedures different from the usual practices of the first-line assessment; (iii) univariable models; (iv) lack of quantitative discrimination power.

Data extraction and synthesis: A data collection form developed by three authors (AB, CGL, PM) was filled by reviewers independently. AB, CGL and PM reviewed the final form. The statistician (AB) and two randomly selected reviewers were charged of data extraction. Disagreements were resolved by consensus. Area under the ROC curve, operative cut-off and related sensitivity and specificity were reported. Heterogeneity of the study findings prevented any meta-analysis.

Results: 5,711 studies were identified and 27 were included. Beside age, sex, and symptom typicality, other common risk factors are smoking, hypertension, diabetes mellitus and dyslipidaemia. Only one model considers genetic profile. Area under the ROC curve values range from 0.51-0.81. Relevant heterogeneity sources have been highlighted, such as sample size, the presence of a cut-off and the adoption of different endpoints. Very few papers address

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a complete validation procedure.

Conclusions: We recommend a more clear statement of endpoints, their consistent measurement both in the derivation and validation phases, more comprehensive validation analyses and the enhancement of threshold validations to assess the effects of pretest models on clinical management.

Strengths and limitation of this study

- This review discusses strengths and weakness of an extensive literature on diagnostic multivariable pretest probability models validated on external populations of subjects suspected of stable coronary artery disease, and quantitatively assessed for discrimination power.
- The multiple sources of heterogeneity highlighted and primarily the various definitions of the outcome, prevented any meta-analysis.
- Given the substantial lack of data on the effect of pretest probability models on the patients' clinical management, this systematic review provides a rational guide to researchers and clinicians as it identifies the best validation practices and provides a set of recommendations for both model development and validation.

Keywords: Coronary Heart Disease; Cardiovascular imaging; Computed tomography; PUBLIC HEALTH

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INTRODUCTION

Cardiovascular Diseases (CVDs) are the leading cause of mortality and morbidity worldwide with 422.7 million prevalent cases and 17.92 million deaths (one-third of all deaths) estimated in the most recent analysis of global burden of CVDs.[1] Coronary artery disease (CAD) accounts for a large proportion of prevalent cases of CVDs after 40 years of age. CAD is one of the important causes of cardiovascular morbidity and mortality with a global estimation of 110.55 million prevalent cases and 8.92 deaths, which makes CAD the leading cause of death in the world.[1]

Stable CAD is most commonly caused by atherosclerotic coronary artery narrowing and is characterized by episodes of reversible myocardial demand/supply mismatch, related to ischaemia or hypoxia, which are usually inducible by exercise, emotion or other stress and commonly associated with transient chest discomfort (stable angina pectoris).[2,3]

Stable CAD diagnosis is established through non-invasive functional and/or anatomical testing,[2,3] and invasive coronary angiography (ICA).[2] Preventive medication plus symptomatic medical management and/or revascularization are the current treatment strategies for established stable CAD.[2,3]

To limit the risk of inappropriate examinations, with its consequences on patients' and healthcare professionals' safety, and economic sustainability of Healthcare Systems,[4–7] eligibility to diagnostic testing is established through models that predict a pre-test probability (PTP) of coronary artery disease (CAD). Since the introduction of the Diamond-Forrester model (DFM)[8] and the Duke Clinical Score (DCS)[9] several alternative PTP models have been proposed and recommended in guidelines for stable symptomatic subjects.[3,10] Recent updates in the European scenario stressed the overestimation flaw of such models. As a consequence, the UK National Institute for Health and Care Excellence (NICE) has preferred a simpler identification of anginal chest pain or abnormal resting electrocardiogram (ECG) as

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a gatekeeper to Coronary computed tomography angiography (CCTA).[11] However, the performance of CCTA for the diagnosis of obstructive CAD is not significantly influenced by chest pain symptoms and angina is more than a mere biological phenomenon which requires specific attentions especially in women.[12,13] The European Society of Cardiology (ESC) updated guideline determines PTP from the stratified prevalence of CAD in a contemporary cohort, instead of recurring to a prediction model as in the past. These new estimated risks are noticeably lower compared to the previous ones possibly suggesting underestimation when applied to different populations, as also recently stated by Bing and colleagues.[14] US Experts are debating on whether adopting the NICE diagnostic approach or keeping on using PTP.[15,16] To face the flaws on available PTP models highlighted by NICE and ESC, these organizations clearly underline the need for more information on the various risk factors acting as modifier of the PTP, especially in the low probability range,[10] and for the development and validation of new scores addressing outstanding uncertainties in the estimation of the PTP of CAD.[11]

This review provides several new contributions to the actual debate on how to ameliorate the PTP models as it focuses on external validation mainly[17] identifies the best results and characterizes the best procedures in terms of significant predictive variables, discriminatory ability and methods completeness. Moreover, the review highlights some key issues that could be further improved in the development and validation phases, to increase decision making capability.

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METHODS

This systematic review conforms to the PRISMA statement;[18] the protocol was registered in PROSPERO (CRD42019139388).[19]

Study inclusion and exclusion criteria

Inclusion and exclusion criteria were developed to identify studies that validated PTP models of stable obstructive CAD (as a binary outcome) anatomically determined through either ICA or CCTA. Reasons of exclusion were: (i) acute coronary syndrome, unstable chest pain, a history of myocardial infarction or previous revascularisation; (ii) models that included a diagnostic procedure that do not reflect the usual practices of the first-line assessment;[3,10] (iii) models based on a single predictive variable; (iv) lack of clearly stated discrimination power. Unlike previous works,[20] external validation was primarily considered. We also included internal validation but limited it to *k*-fold cross-validation as a technique inspired by the same purposes of external validation. Moreover, papers referring to Machine learning-based PTP models have been excluded as considered in a recent review focusing on CAD diagnosis by ML with aims close to ours.[21]

Only full papers were retained because other publications, e.g., letters to editors, conference proceedings, et cetera, are usually not assessed for study quality. Only articles published in English and Italian were considered.

Searches

The databases Global Health, Healthstar and MEDLINE® were systematically searched (CGL, PM) on 22 April 2020 using several keywords, including: angina pectoris, chest pain, coronary artery disease, coronary heart disease, coronary stenosis, stratification score, likelihood function, predictive model, pre-test probability, coronary angiography, cardiac catheterisation and computed tomography angiography. Full electronic search strategy is reported in Additional file 1. Citation searches were also performed on reference lists of definitively

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included studies.

Study selection

A multidisciplinary working team was composed. Eligibility screening was performed independently in an unblinded standardized manner by all the reviewers. Preliminary screening was performed using Abstrackr[22] based on title and abstract with each paper assessed by two randomly assigned reviewers. Selected papers were assessed based on full text. Disagreements were resolved by consensus.

Data extraction strategy

A data collection form was developed by three authors (AB, CGL, PM) and filled by reviewers independently. Three authors (AB, CGL, PM) reviewed the final form for internal consistency. Each selected paper was assigned for data extraction to the statistician (AB) and two randomly selected reviewers.

Study quality assessment

The quality assessment of included studies conforms to QUADAS-2 and was performed by four reviewers (AB, CGL, PM, MRT).[23] Due to the previously described features (i)-(iv), we considered that the eligible works did not raise applicability concerns.

Data synthesis and presentation

The performances of prediction models can be summarised using several methods and indices, and the area under receiver operating characteristic curve (AUC) is certainly the best-known. Sensitivity and specificity also describe the discrimination capability of the model for a given cut-off and thus provide an indication of clinical usefulness.

For the purposes of generalisation of a PTP model to populations that differ from the development population study, the computation of performance indexes is not sufficient because a lower performance is usually expected.[17,24] Therefore, we also noted whether more extended validation procedures were performed in order to properly apply a model to

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new populations.

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VALIDATED PTP MODELS: MAIN RESULTS

Study selection

A total of 5,711 studies were identified (3 through reference lists of included studies) and 2,685 different abstracts were screened. Out of the 71 relevant full-texts assessed for eligibility, 27 were finally included (Figure 1).

Study characteristics

Table 1 summarises the selected studies in terms of model name, geographical location, population recruitment criteria. Sometimes the same model is referenced with different names across the papers, then Table 1 indicates the original name and the one we adopted here.

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Table 1 Characteristics of the studies on PTP for CAD

Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Adamson PD, 2018a[25]	DFM/CASS uDFM	1) Multicenter PROMISE trial, US & Canada 2) Multicenter SCOT-HEART trial, Scotland (UK)	See PROMISE. Randomised to receive CCTA as non-initial non-invasive test. See SCOT-HEART. Randomized to the CCTA intervention arm.	See PROMISE & SCOT-HEART. Known CAD
Adamson PD, 2018b[26]	uDFM (Baseline CADC model, in text) uDFM-cTn (Baseline CADC model with the addition of troponin, in text)	Odense University Hospital, Denmark	Clinical stable prospectively enrolled patients with suspected angina pectoris scheduled for either ICA or CCTA[27]	Suspected acute coronary syndrome. To avoid potential confounding effects on the biomarkers measured, patients with established atherosclerotic manifestations, including an abnormal 12-lead rest electrocardiogram, were excluded: known ischemic heart disease, prior ischemic stroke or transitory ischemic attack, known peripheral artery disease (n = 10), and p-creatinine >200 mmol/L. CCTA not performed or of poor technical quality, lack of informed consent, missing hs-cTnI measure or personal history.[27]

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Almeida J, 2016[28]	CADC-Clin (CAD Consortium 2, in text) DCS uDFM (CAD Consortium 1, in text)	Single center in Southwestern Europe	Patients with chest pain and suspected CAD referred to ICA	Patients with a history of CAD, acute coronary syndrome, or coronary revascularization
Baskaran L, 2018[29]	CADC-Clin CONFIRM score uDFM	Multicenter SCOT-HEART trial, Scotland (UK)	See SCOT-HEART. Randomized to the CCTA intervention arm and with information on all variables needed for the analysis.	See SCOT-HEART. Known CAD
Bittencourt MS, 2016[30]	CADC-Basic CADC-Clin uDFM (Diamond and Forrester score, in text)	Massachusetts General Hospital; Brigham and Women's Hospital (Massachusetts, USA)	Subjects ≥ 18 years who underwent CCTA for suspect of CAD	Patients who were missing any of the clinical information needed to calculate the pretest probability, who had nondiagnostic CCTA images, who had incomplete follow-up information; with congenital heart disease, heart transplantation, or prior CAD, defined as prior percutaneous coronary interventions, coronary artery bypass graft surgery, or MI

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Daniels SE, 2014[31]	Corus® CAD (Gene Expression score – GES, in text)	Multicenter PREDICT trial US	See PREDICT	See PREDICT. Diabetic patients
Edlinger M, 2017[32]	CADC-Clin	University Clinic of Cardiology at Innsbruck (Austria)	Patients were 18 years of age or older with chest pain or symptoms suggestive of CAD (predominantly dyspnoea) and/or non-invasive evidence of CAD referred for elective ICA.	1) an elective ICA before or after heart transplantation, 2) an elective ICA prior to solid organ transplantation, 3) an elective ICA before heart valve repair or replacement, or with valvular heart disease as leading clinical diagnosis, 4) an isolated right heart catheterisation, 5) an electrophysiological procedure (pace-maker implantation or catheter ablation) as leading clinical indication, 6) an elective ICA because of a known or suspected congenital heart disease as leading clinical diagnosis (e.g., atrial septal defect, ventricular septal defect or patent foramen ovale), or 7) when referred for other reasons (like myocardial biopsy, aortic aneurysms, myxoma, endocarditis or prior failed angiography). History of myocardial infarction.

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Ferreira AM, 2016[33]	uDFM (Modified DF, in text) CADC-Clin (CAD consortium 2, in text) CONFIRM score	Unspecified, Portugal	Patients undergoing CCTA for the evaluation of CAD	Age <30 years; known CAD; suspected acute coronary syndrome; preoperative assessment; known left ventricular systolic dysfunction; asymptomatic patients (typically referred after a positive screening exercise test); symptoms other than chest pain. Patients with suspected CAD who were scheduled to undergo CCTA but had the procedure halted due to a high coronary artery calcium (CAC) Agatston score. A threshold of 400 was used as a general guideline for withholding CCTA in these circumstances, but the decision was ultimately left to the performing physician, taking into consideration the clinical context and the distribution of calcium in the coronary tree.
Fordyce CB, 2017[34]	PROMISE minimal risk model (The originally published version has been subsequently corrected	Multicenter PROMISE trial, US & Canada	See PROMISE. Patients assigned to anatomic testing	See PROMISE

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
	online, see Fordyce CB, 2018[35])			
Fujimoto S, 2014[36]	DCS K-score	Multicenter, Japan	Suspected CAD	Patients with known CAD, showing poor image quality and patients with un-assessable segments due to severe calcification
Genders TSS, 2011[37]	DFM	14 European centers	Patients aged 30-69 with stable chest pain (typical, atypical, or non-specific chest pain) and if ICA performed.	Patients meeting the following criteria: (i) acute coronary syndrome or unstable chest pain, (ii) history of myocardial infarction or previous revascularization (percutaneous coronary intervention or coronary artery bypass graft surgery), and (iii) no informed consent.
	uDFM	Erasmus Medical Center, Rotterdam, the Netherlands[38]	Patients with stable chest pain and no history of CAD[38]	Not undergoing CCTA or ICA
Genders TSS, 2012[39]	DCS	Multicenter EU and US	Stable chest pain, referred for catheter based or CT based coronary angiography	Acute coronary syndrome, unstable chest pain, history of myocardial infarction or previous revascularization or no informed consent.

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
	CADC-Basic CADC-Clin	Multicenter EU and US	Stable chest pain, referred for catheter based or CT based coronary angiography	Acute coronary syndrome, unstable chest pain, history of myocardial infarction or previous revascularization or no informed consent.
Genders TSS, 2018[40]	CADC-Basic CADC-Clin	Multicenter PROMISE trial, US & Canada	See PROMISE Trial for the main criteria. Patients assigned to anatomic testing	See PROMISE Trial for the main criteria
Jensen JM, 2012[41]	CORSCORE DCS DFM Morise score uDFM	Lillebælt Hospital Vejle, Denmark	Patients with chest pain indicative of CAD referred for ICA	Unstable angina or previous coronary intervention
Min JK, 2015[42]	CONFIRM score (Integer-based risk model, in text)	United States, Canada, South Korea and Austria (4 out of 5 sites of the Phase II of	Patients ≥18 years old referred to CCTA for suspected stable CAD (CONFIRM trial[43])	Patients with prior coronary revascularization or MI, asymptomatic, missing data

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
		CONFIRM trial[43])		
Pickett CA, 2013[44]	DFM/CASS Morise score	Walter Reed Army Medical Center, Washington USA	Patients referred for CCTA	Known CAD
Rademaker AA, 2014[45]	DCS DFM Morise score (New score, in text) uDFM	VU University Medical Center, Amsterdam, The Netherlands	Symptomatic women undergoing evaluation for CAD and referred for CCTA	Prior history of CAD (percutaneous coronary intervention, coronary artery bypass graft surgery, or previous myocardial infarction), or absolute or relative contraindications for CCTA such as (i) significant severe arrhythmia; (ii) pregnancy; (iii) renal insufficiency (glomerular filtration rate<45 ml/min); (iv) known allergy to iodinated contrast material.
Rosenberg S, 2010[46]	Corus® CAD (Gene expression test, in text) Expanded clinical model score	Multicenter PREDICT trial US	See PREDICT	See PREDICT. Diabetes

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
	DFM/CASS			
Teressa G, 2018[47]	CADC-Basic CADC-Clin	1 center in US	>18 years old evaluated in the Emergency Department of a major academic tertiary university hospital for chest pain, using CCTA as a primary diagnostic modality	Known CAD, defined as history of acute myocardial infarction, percutaneous intervention, coronary artery bypass graft, or evidence of CAD by either anatomical (CCTA or cardiac catheterization) or functional tests (positive stress test). Hemodynamically or clinically unstable patients, patients with ST segment changes or positive cardiac troponin (>0.04ng/ml), impaired renal function (eGFR<50ml/min/1.73m^2), tachycardia, or contraindication to nitroglycerin or iodinated contrast. Inadequate documentation on Chest pain characteristics, repeat CCTAs, unavailable calcium score and non-diagnostic exam.
Thomas GS 2013[48]	Corus® CAD (GES, in text) DFM Morise score	Multicenter COMPASS trial, US	See COMPASS	See COMPASS

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Voora D, 2017[49]	Corus® CAD	Multicenter PROMISE trial, US & Canada	See PROMISE. Patients assigned to anatomic testing	See PROMISE. Diabetes. RNA sample not passing quality control.
Voros S, 2014[50]	Corus® CAD (GES, in text) DFM	Multicenter PREDICT US and COMPASS US trials	See PREDICT and COMPASS.	See PREDICT and COMPASS. Diabetes excluded from PREDICT cohort.
Wang M 2018[51]	CONFIRM score	Not specified, China	Patients who underwent CCTA for stable chest pain and with 0 or 1 risk factors among smoking, hypertension, diabetes and hyperlipidemia	Acute coronary syndrome, previous CAD or coronary revascularization, un-assessable segments due to motion artifact, atrial fibrillation, aortic disease, New York Heart Association class III or IV heart failure, age > 90 years old, pacemaker leads or missing data
Winther S, 2019[52]	uDFM CADC-Basic CADC-Clin	Multi-center Dan-NICAD trial, Denmark	Patients without known CAD referred to CCTA due to a history of symptoms suggestive of CAD	Age <40; previous coronary revascularization or MI; unstable angina pectoris; estimated glomerular filtration rate <40mL/min; pregnancy; and contraindication for iodine-containing contrast medium, magnetic resonance imaging, or adenosine (severe asthma, advanced atrioventricular block, or critical aortic stenosis).

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Yang Y, 2015[53]	High Risk Anatomy (HRA) score	Multicenter CONFIRM trial,[43] North America, Europe and Asia University of Ottawa Heart Institute Cardiac CT registry	Patients ≥18 years old referred to CCTA for suspected stable CAD (CONFIRM trial)[43]	Documented CAD, history of MI, coronary revascularization, cardiac transplantation, congenital heart disease
	uDFM	Multicenter CONFIRM trial,[43] North America, Europe and Asia	Patients ≥18 years old referred to CCTA for suspected stable CAD (CONFIRM trial)[43]	Documented CAD, history of MI, coronary revascularization, cardiac transplantation, congenital heart disease

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Zhang Y, 2019[54]	DCS uDFM	Tianjin Chest Hospital, Tianjin, China	Patients with stable chest pain and referred for CCTA	Acute coronary syndrome, previous CAD or coronary revascularization (percutaneous coronary intervention or coronary artery bypass grafting), impaired renal function (serum creatinine > 120 μmol/l), New York Heart Association class III or IV heart failure, atrial fibrillation, aortic disease, age more than 90 years, or patients with un-assessable segments because of artefact
Zhou J, 2017[55]	CADC-Clin (Genders clinical model, in text) DCS uDFM	Not specified, China	Patients who underwent CCTA for stable chest pain	Acute coronary syndromes, previous CAD or coronary revascularization (percutaneous coronary intervention or coronary artery bypass grafting), patients with un-assessable segments due to motion artefact, atrial fibrillation, aortic disease, New York Heart Association class III or IV heart failure, age > 90 years, presence of pacemaker leads or missing data.

The trials COMPASS, CONFIRM, PREDICT, PROMISE and SCOT-HEART were considered in several studies, and thus their main characteristics are fully reported in Additional File 2

Legend

CADC-Basic = CAD Consortium Basic model

CADC-Clin = CAD Consortium Clinical model

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CASS = Coronary Artery Surgery Study

DCS = Duke Clinical Score

DFM = Diamond & Forrester (DF) model

eGRF = estimated glomerular filtration rate

NA = Not Available

QCA = Quantitative Coronary Angiography

uDFM = Updated Diamond-Forrester model

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Studies are mainly conducted in North America [30,31,34,40,44,46–50] or Europe[26,28,29,32,33,37,41,45,52].

The Updated Diamond-Forrester model (uDFM),[25,26,53–55,28–30,33,37,41,45,52] and the CAD consortium clinical model (CADC-Clin)[28–30,32,33,39,40,47,52,55] are the most assessed models.

The quality of included studies is generally high due to the specific review question and adopted eligible criteria. Nevertheless, risk of bias arises from a few specific issues. A few validation studies[31,37,39,46,50] do not declare that they enrolled only consecutive or random samples of patients. With respect to the index test, only one work adopted an optimal discriminating threshold in addition to pre-specified ones.[50] Application of CCTA as a reference test yields a risk of bias in many studies[25,34,37,39,40,42,45,49,53] that do not report measures against misclassification of the test results. Finally, in four works[26,39,40,48] patients did not receive the same reference test for the diagnosis of stable CAD. A graphical summary of the risk of bias is reported in Additional File 3.

Predictive variables and discrimination capability

As shown in Table 2, the identified models can be classified into two broad classes: basic models, including the DFM (based on age, sex and chest pain) and its updates, and clinical models, including the DCS and the models that extend the DFM by adding a few, mainly traditional,[56] risk factors. Within this quite classical framework, the Corus® CAD model is distinguished by relating CAD in nondiabetic patients to the expression levels of a set of genes.

All the models were derived by logistic regression. Exceptions are: DFM, derived by a conditional probability analysis in the late 1970s; Corus® CAD, obtained through Ridge regression; CONFIRM score, developed to predict adverse clinical events by fitting a Cox proportional hazards model and subsequently validated for diagnosis of CAD.

Cross-validation[39] and split-sample[34,46] have been used in a few cases only.

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Table 2 PTP models' variables

Macro categories	Predicting variables	Model/Score															
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
		[30,39,40,47,52]	[28–30,32,33,39,40,47,52,55]	[29,33,42,51]	[41]	[31,46,48–50]	[28,36,39,41,45,54,55]	[37,41,42,45,48,50]	[25,44,46]	[46]	[36]	[53]	[41,44,45,48]	[34]	[25,26,53–55,28–30,33,37,41,45,52]	[26]
Demography	Age	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	Sex	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	Race									√				√		
Medical history	Diabetes mellitus		√	√			√				√	√	√	√		
	Hypertension		√	√	√					√	√	√	√	√		

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Macro categories	Predicting variables	Model/Score															
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
	Previous MI				√		√									
	Cerebral Infarction										√					
	Peripheral vascular disease											√				
Clinical presentation/ physical examination	Chest pain	√	√	√	√		√	√	√	√	√	√	√		√	√
	Abnormal ECG						√									
	Obesity												√			
	Smoking		√	√	√		√				√	√	√	√		

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Macro categories	Predicting variables	Model/Score														
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
	Family history of CAD			√								√	√	√		
	Other (specify)				Medically treated hypercho l.					Medically treated hypercho l.				Symptoms related to physical or mental stress		
Bio-Chemistry	HDL cholesterol													√		
	Dyslipidaemia		√				√			√	√	√	√	√		

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Macro categories	Predicting variables	Model/Score														
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
	Oestrogen status												√			
	Gene expression					√										
	Troponin															√
Others										Aspirin, anti-platlet, ACE inhibitor use, systolic						

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Macro categories	Predicting variables	Model/Score														
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
										blood pressure						
Derivation method		Log	Log	Cox proportional hazards models	Log	Score derived by a Ridge regression	Log	Conditional probability analysis*	Log	Log	Log	Score derived by a multivariable Log	Score derived by a Log	Log	Log	Log

* In Genders, 2011,[37] to unravel the implicit coefficients of the predictors in this model, the authors performed a weighted linear regression on the log odds of the DF predictions per subgroup

Legend

DCS = Duke Clinical Score

DFM = Diamond & Forrester model

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Log = Logistic regression

QCA = Quantitative Coronary Angiography

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Predictors were classified into four macro-areas: demography, medical history, clinical presentation/physical examination and biochemistry. The demographic macro-area is present in all models with the variables age and sex, while race is only included in the Expanded clinical model and PROMISE Minimal Risk model. The most used variables in the medical history macro-area are diabetes mellitus and hypertension. The clinical presentation/physical examination macro-area is present in all but the Corus® CAD models. Only the Corus® CAD and PROMISE Minimal Risk models do not include chest pain. The most used variable in the biochemistry macro-area is dyslipidaemia. The other risk factors are model-specific: gene expression (Corus® CAD), oestrogen status (Morise score), high-density lipoprotein cholesterol (PROMISE Minimal Risk model) and the high-sensitivity cardiac troponin (uDFM-cTn).

Finally, Table 3 reports the overall picture of the PTP discriminatory abilities in the validation studies.

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Table 3 PTP Models performance as reported in the selected studies

Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
CADC-Basic	Bittencourt MS, 2016[30]	At least 1 segment (with a >2 mm diameter) with a lesion with ≥50% diameter stenosis	CCTA	2,274	22	0.7517 (0.729 - 0.775)	*0.62* (5%)	*94.0*	*29.9*
	Genders TSS, 2012[39]	≥1 diameter stenosis of ≥50% in ≥1 vessel	CCTA, ICA	min: 471 max: 1,241	NA	Mean: 0.77			
	Genders TSS, 2018[40]	≥1 diameter stenosis of ≥50% in ≥1 vessel (≥2.0 mm diameter) by ICA. Patients with a completely normal CCTA (0% stenosis and coronary artery calcium score of 0) are considered as free of obstructive CAD on ICA.	CCTA, ICA	3,468	23	0.69 (0.67 - 0.72)	*0.62* (10%)	83 (81 - 86)	40 (38 - 41)
	Teressa G, 2018[47]	1 vessel with stenosis of 50%	CCTA	1,981	10.4	0.77 (0.73 - 0.77) sic in the text	*0.66* (5%)	*85.4*	*46.8*

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
							0.69 (10%)	*67.5*	*69.6*
							0.65 (20%)	*39.8*	*89.9*
	Winther S, 2019[52]	Coronary diameter stenosis reduction ≥50% in all segments with a reference vessel diameter >2mm	CCTA	1,653	23.7	0.66 (0.63–0.69)			
CADC-Clin	Almeida J, 2016[28]	Stenosis of >50% in at least one major epicardial vessel	ICA	2,234	58.5	0.683 (0,661 - 0.706)	*0.60* (15%)	*91.3*	*27.8*
	Baskaran L, 2018[29]	A stenosis causing ≥50% diameter stenosis	CCTA	1,738	37.7	0.790 (0.768 - 0.811)			
	Bittencourt MS, 2016[30]	At least 1 segment (with a >2 mm diameter) with a lesion with ≥50% diameter stenosis	CCTA	2,274	22	0.791 (0.770 - 0.812)	*0.67* (5%)	*95.8*	*37.3*

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Edlinger M, 2017[32]	Stenosis ≥50% diameter in at least one of the main coronary arteries	ICA	4,888	44	0.69 (0.67 - 0.70)			
	Ferreira AM, 2016[33]	Coronary diameter stenosis ≥50%	CCTA	1,069	13.8	0.73 (0.71 - 0.76)			
	Genders TSS, 2012[39]	≥1 diameter stenosis of ≥50% in ≥1 vessel	CCTA, ICA	min: 471 mean: NA max: 1,241	NA	0.78 0.79 0.81			
	Genders TSS, 2018[40]	≥1 diameter stenosis of ≥50% in ≥1 vessel (≥2.0 mm diameter) by ICA. Patients with a completely normal CCTA (0% stenosis and coronary artery calcium score of 0) are considered as free of obstructive CAD on ICA.	CCTA, ICA	3,468	23	0.72 (0.69 - 0.74)	*0.63* (10%)	89 (87 - 91)	36 (34 - 38)
	Teressa G, 2018[47]	1 vessel with stenosis of 50%	CCTA	1,981	10.4	0.80 (0.77 - 0.80)	*0.69* (5%)	*86.4*	*51.3*

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
						sic in the text			
							0.72 (10%)	*71.4*	*72.7*
							0.67 (20%)	*45.6*	*88.7*
	Winther S, 2019[52]	Coronary diameter stenosis reduction ≥50% in all segments with a reference vessel diameter >2mm	CCTA	1,653	23.7	0.69 (0.66–0.72)			
	Zhou J, 2017[55]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	5,743	32.6	0.774 (0.761 - 0.788)			
CONFIRM score	Baskaran L, 2018[29]	A stenosis causing ≥50% diameter stenosis	CCTA	1,738	37.7	0.749 (0.726 - 0.771)			
	Ferreira AM, 2016[33]	Coronary diameter stenosis ≥50%	CCTA	1,069	13.8	0.71 (0.66 - 0.75)			

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Min JK, 2015[42]	≥50% luminal diameter stenosis in any coronary artery ≥1.5 mm in diameter	CCTA	2,132	NA	0.76 (0.746 - 0.771)			
	Wang M 2018[51]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	0 Risk Factors (RF): 1,201	30.2	0.756 (0.731 - 0.781)			
				1 RF: 2,415	27.1	0.762 (0.742 - 0.783)			
CORSCORE	Jensen JM, 2012[41]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.727			
Corus® CAD	Daniels SE, 2014[31]	At least one lesion in a major coronary artery (≥1.5 mm lumen diameter) ≥70% diameter stenosis by clinical read or ≥50% diameter stenosis by invasive QCA	ICA	Several subsets from a total of 1,502	NA	min: 0.64 max: 0.72			
	Rosenberg S, 2010[46]	≥1 atherosclerotic plaque in a major coronary artery (≥1.5 mm lumen diameter) causing ≥50% luminal diameter stenosis by QCA	ICA	526	36.5	0.70 (0.68 - 0.72)	*0.64* (14.75,	*84.9*	*43.5*

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
							range 0 to 40)		
	Thomas GS 2013[48]	≥1 diameter stenosis ≥50% in a major vessel on ICA by QCA (≥1.5 mm) or CCTA (≥2.0 mm)	CCTA, ICA	431	14.6	0.79 (0.72 - 0.84)	*0.71* (15)	89 (78 - 95)	52 (47 - 57)
	Voora D, 2017[49]	≥70% stenosis in major coronary artery or ≥50% left main stenosis	CCTA	1,137	10.1	0.625 (0.573 - 0.678)	*0.60* (15)	*73*	*47.7*
	Voros S, 2014[50]	Outcome 50: ≥ 50% maximum diameter stenosis	CCTA	610	14 PREDICT (16)	0.75 (0.70 - 0.80)	*0.68* (15)	84	51
							0.69 (19, Optimal)	76	61
					COMPASS (13)		*0.65* (28)	40	90
		Outcome 70: ≥ 70% maximum diameter stenosis	CCTA		NR	0.75 (0.67 - 0.83)	*0.69* (15)	90	48

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
							0.71 (19, Optimal)	84	58
							0.66 (28)	45	87
DCS	Almeida J, 2016[28]	Stenosis of >50% in at least one major epicardial vessel	ICA	2,234	58.5	0.685 (0.663 - 0.708)	*0.54* (15%)	*98.0*	*9.9*
	Fujimoto S, 2014[36]	Lesions with diameter stenosis of ≥75% were defined to be obstructive stenotic lesions. As for left main trunk lesion, lesions with diameter stenosis ≥50% were defined to be obstructive stenotic lesions.	CCTA	361	34.1	0.688 (0.626 - 0.750)			
	Genders TSS, 2012[39]	Severe CAD defined as ≥70% diameter stenosis or ≥50% left main stenosis	CCTA, ICA	4,426	NA	0.78 (0.76 - 0.81)			
	Jensen JM, 2012[41]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.718			

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Rademaker AA, 2014[45]	>50% luminal diameter stenosis	CCTA	178	23.6	0.59 (0.51 - 0.66)			
	Zhang Y, 2019[54]	≥1 lesion with ≥50% diameter stenosis	CCTA	Men: 3,001	39	0.785	*0.73* (15%)	*92.6*	*52.4*
				Women: 2,776	25	0.684	*0.73* (15%)	*80.7*	*64.6*
	Zhou J, 2017[55]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	5,743	32.6	0.772 (0.759 - 0.786)			
DFM	Genders TSS, 2011[37]	≥50% diameter stenosis in ≥1 vessel	ICA	1,683	55.7	0.78 (0.76 - 0.79)			
	Jensen JM, 2012[41]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.642			

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Min JK, 2015[42]	≥50% luminal diameter stenosis in any coronary artery ≥1.5 mm in diameter	CCTA	2,132	NA	0.64 (0.628 - 0.659)			
	Rademaker AA, 2014[45]	>50% luminal diameter stenosis	CCTA	178	23.6	0.56 (0.49 - 0.64)			
	Thomas GS 2013[48]	≥1 diameter stenosis ≥50% in a major vessel on ICA by QCA (≥1.5 mm) or CCTA (≥2.0 mm)	CCTA, ICA	431	14.6	0.69 (0.62 - 0.75)			
	Voros S, 2014[50]	Outcome 50: ≥ 50% maximum diameter stenosis	CCTA	610	14 PREDICT (16%) COMPASS (13%)	0.65 (0.59 - 0.71)			
		Outcome 70: ≥ 70% maximum diameter stenosis	CCTA		NR	0.63 (0.53 - 0.73)			

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
DFM/CASS	Adamson PD, 2018a[25]	≥70% area stenosis in any major epicardial vessel or ≥50% stenosis in the left main stem	CCTA	4,541 (PROMISE)	11.8	0.510 (0.506 - 0.514)	*0.51* (10%)	99.4 (98.4 - 99.9)	2.6 (2.2 - 3.2)
			CCTA	1,619 (SCOT-HEART)	22.2	0.560 (0.548 - 0.573)	*0.56* (10%)	97.5 (96.6 - 98.1)	14.6 (13.0 - 16.4)
	Pickett CA, 2013[44]			1,027	6.82	0.72 (0.66 - 0.78)			
	Rosenberg S, 2010[46]	≥1 atherosclerotic plaque in a major coronary artery (≥1.5 mm lumen diameter) causing ≥50% luminal diameter stenosis by QCA	ICA	526	36.5	0.663 (0.638 - 0.688)			
Expanded Clinical Model	Rosenberg S, 2010[46]	≥1 atherosclerotic plaque in a major coronary artery (≥1.5 mm lumen diameter) causing ≥50% luminal diameter stenosis by QCA	ICA	526	36.5	0.732	*0.62* (20%)	*85.6*	*38.0*

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
HRA score	Yang Y, 2015[53]	High-risk CAD: left main coronary artery diameter stenosis ≥50%, 3-vessel disease (≥70%), or 2-vessel disease involving the pLAD artery	CCTA	7,333	4.8	0.71 (0.69 - 0.74)	*0.51* (18, range -1 to 25)	2.3	99.4
K-score	Fujimoto S, 2014[36]	Lesions with diameter stenosis of ≥75% were defined to be obstructive stenotic lesions. As for left main trunk lesion, lesions with diameter stenosis ≥50% were defined to be obstructive stenotic lesions.	CCTA	361	34.1	0.712			
Morise score	Jensen JM, 2012[41]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.681			
	Pickett CA, 2013[44]	≥50% visual luminal diameter stenosis in ≥1 epicardial coronary artery segment ≥1.5 mm in diameter	CCTA	1,027	6.82	0.68 (0.63 - 0.74)			

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Rademaker AA, 2014[45]	>50% luminal diameter stenosis	CCTA	178	23.6	0.67 (0.60 - 0.74)			
	Thomas GS 2013[48]	≥1 diameter stenosis ≥50% in a major vessel on ICA by QCA (≥1.5 mm) or CCTA (≥2.0 mm)	CCTA, ICA	431	14.6	0.65 (0.59 - 0.74)			
PROMISE Minimal Risk model	Fordyce CB, 2017[34]	Minimal risk: normal CCTA and further conditions*	CCTA	1,528	25.0	0.713 (0.684 - 0.742)			
uDFM	Adamson PD, 2018a[25]	≥70% area stenosis in any major epicardial vessel or ≥50% stenosis in the left main stem	CCTA	4,541 (PROMISE)	11.8	0.510 (0.506 - 0.514)	*0.51* (15%)	99.4 (98.4 - 99.9)	2.6 (2.2 - 3.2)
			CCTA	1,619 (SCOT-HEART)	22.2	0.594 (0.579 - 0.610)	*0.59* (15%)	95.8 (94.7 - 96.7)	23.0 (21.0 - 25.1)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Adamson PD, 2018b[26]	Luminal cross-sectional area stenosis of ≥70% (approximating to a 50% diameter stenosis) in at least 1 major epicardial vessel or ≥50% in the left main stem.	CCTA, ICA	487	19.3	0.738 (0.687 - 0.788)			
	Almeida J, 2016[28]	Stenosis of >50% in at least one major epicardial vessel	ICA	2,234	58.5	0.664 (0.641 - 0.687)	*0.52* (15%)	*99.5*	*3.8*
	Baskaran L, 2018[29]	A stenosis causing ≥50% diameter stenosis	CCTA	1,738	37.7	0.767 (0.744 - 0.790)			
	Bittencourt MS, 2016[30]	At least 1 segment (with a >2 mm diameter) with a lesion with ≥50% diameter stenosis	CCTA	2,274	22	0.714 (0.689 - 0.737)	*0.54* (5%)	*98.2*	*10.1*
	Ferreira AM, 2016[33]	Coronary diameter stenosis ≥50%	CCTA	1,069	13.8	0.70 (0.67 - 0.72)			

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
	Genders TSS, 2011[37]	≥50% diameter stenosis in ≥1 vessel	ICA	471	NA	0.76 (0.71 - 0.81)			
	Jensen JM, 2012[41]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.714			
	Rademaker AA, 2014[45]	>50% luminal diameter stenosis	CCTA	178	23.6	0.61 (0.53 - 0.68)			
	Winther S, 2019[52]	Coronary diameter stenosis reduction ≥50% in all segments with a reference vessel diameter >2mm	CCTA	1,653	23.7	0.65 (0.61– 0.68)			
	Yang Y, 2015[53]	High-risk CAD: left main coronary artery diameter stenosis ≥50%, 3-vessel disease (≥70%), or 2-vessel disease involving the pLAD artery	CCTA	24,251	3.6	0.64 (0.62 - 0.67)			
	Zhang Y, 2019[54]	≥1 lesion with ≥50% diameter stenosis	CCTA	Men: 3,001	39	0.782	*0.65* (15%)	*79.6*	*50.2*

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)	AUC (cut-off)	Sensitivity (95% CI) [%]	Specificity (95% CI) [%]
				Women: 2,776	25	0.678	*0.67* (15%)	*91.1*	*42.8*
	Zhou J, 2017[55]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	5,743	32.6	0.765 (0.751 - 0.779)			
uDFM-cTn	Adamson PD, 2018b[26]	Luminal cross-sectional area stenosis of ≥70% (approximating to a 50% diameter stenosis) in at least 1 major epicardial vessel or ≥50% in the left main stem.	CCTA, ICA	487	19.3	0.757 (0.706 - 0.808)			

* Further conditions are considered and should be all present, in addition to normal CCTA, for a subject to be at minimal risk: (1) coronary artery calcium score was 0 or was not obtained; (2) no evidence of atherosclerosis; (3) overall study quality was diagnostic (i.e., sufficient data quality for interpretation); (4) left ventricular function was normal or not reported; (5) no wall motion abnormalities were present or not reported; and (6) no relevant cardiovascular incidental findings that could account for the patients' symptoms (i.e., aortic dissection or pulmonary embolism) were noted. All patients with normal CCTA results were included in the minimal-risk cohort in the absence of any of the following adjudicated clinical events during the median 25-month follow-up period: all-cause death, nonfatal MI, unstable angina hospitalization, or revascularization during the entire follow-up period

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Legend

Values in Italic are derived by reviewers (AB, CGL, PM)

NA = Not Available

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It is worth noting the presence of relevant heterogeneity sources: the high variability of sample sizes (from 178[45] to 24,251[53]); whether the AUC is computed with respect to a specific PTP cut-off; the adoption of different endpoints. On the one hand, Fordyce et al.[34] focused on patients unlikely to have CAD, clinical events or revascularisation, who were defined as being at "minimal risk". Minimal risk is characterised by a normal CCTA and the presence of additional positive conditions. On the other hand, Yang et al.[53] consider subjects with a high-risk CAD, defined as left main coronary artery diameter stenosis ≥ 50%, 3-vessel disease (diameter stenosis ≥ 70%) or 2-vessel disease involving the proximal left anterior descending coronary artery. Endpoint heterogeneity is also one of the reasons for the large variation of prevalence, from 3.6%[53] to 58.5%.[28]

AUC values range from 0.51[25] to approximately 0.81[39]. These data indicate a degree of discriminative performance that varies from almost failing to almost excellent. The CAD-Clin model only has an AUC > 0.80, and this performance level is confirmed in other validations (AUC≥0.79 and 95% confidence intervals [CIs] including 0.80).[29,30,47] The external validations for the American College of Cardiology Foundation (DFM/CASS model) and ESC guidelines (uDFM)[25] yield the lowest AUC. This result is not unexpected because these values arise from considering the recommended discriminating cut-offs (10% and 15%, respectively) that directly reflect the preference for high sensitivity. These values cannot be compared to other AUC values that are not derived from fixed cut-off: for ease of comparison, in Table 3 the distinction is made between running and fixed cut-off. All the other models except the Morise score (AUC from 0.65[42] to 0.68[32,50]) reached a moderate discriminative ability (AUC from 0.70-0.80) when considering the running cut-off AUCs.

The uDFM has been validated on a very different sized populations (from 173 to more than 20,000 subjects) with variable prevalence (from 3.6-58.6%). The most complete validation of the model, considering calibration-in-the-large, recalibration and eventually re-estimation, has been performed by the developers themselves[37] who obtained a valid overall effect of predictors. The other validating procedures limit themselves to AUC computation and to a rough assessment of under/overestimation, mainly by the Hosmer-Lemeshow goodness-of-fit (HL) test and related

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calibration plots (calibration-in-the-large is applied in one study[33]). The extension of uDFM with the use of high-sensitivity cardiac troponin I (uDFM-cTn), has a significantly higher AUC than uDFM alone (0.757 versus 0.738, p= 0.025) and better calibration HL p= 0.0001 versus HL p= 0.1123).[25] The uDFM updated and extended the traditional DFM to a contemporary cohort that included subjects 70 years and older. The CAD Consortium Basic model (CADC-basic) can be considered as a further update on a different contemporary population (see Table 2). With regards to the DFM (and its DFM/CASS version), overestimation is usually reported, especially in women.[45] Apart from one study,[37] the DFM was not extensively validated but only used as a usual reference model[41,44,45] or as a basis to establish the performances of the Corus® CAD model.[46,48,50]

Among the models that include clinical risk factors, DCS and CADC-Clin have been extensively validated. The former generally overestimates prevalence and shows a lack-of-fit by the HL test. Moreover, miscalibration results from a reduced effect of sex and chest pain typicality and an increased effect of diabetes and dyslipidaemia.[39] The latter has been verified by external validation[32,40,47,52] Results on miscalibration analysis could be considered quite consistent across papers. This finding indicates smaller than expected effects of the diagnostic characteristics, chest pain typicality in particular.[32,40,47] Model calibration can be worse in women compared to men, a situation that also arises from the validation of other models (e.g., DFM[37]). The CADC-Clin performances significantly improve with respect to the related CADC-basic.[30,39,40,47] Comparisons of either uDFM or CADC-Clin with the PROMISE history-based score do not lead to a clear evaluation of the advantages of one over the other in terms of AUC,[29,33] while the CONFIRM score proves to be better than the DFM.[42] The substantially steady results of the CONFIRM score on several data-sets are also confirmed on a validation data-set consisting of subjects at the low extreme of traditional cardiovascular risk factor burden.[51]

The Corus® CAD model stands out from the previous ones because it defines an age- and sex-specific gene expression score. Validation is performed by AUC comparisons, HL test and additivity to DFM and other models. The validation procedures show significant AUC improvement when the score is

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added to other models (e.g., 0.81 versus 0.65 when added to Morise score, with non-overlapping confidence intervals[48]; 0.721 versus 0.663 when added to DFM, p= 0.003[46]; not shown in the table). With respect to the Morise score, the only model that explicitly considers a female-specific factor, namely the oestrogen status, the Corus® CAD has significantly higher AUC (0.79 versus 0.65, p< 0.001[48]). Testing the Corus® CAD model on different data sets from an extension of the original validation population provides results very similar to the original ones.[31]

Finally, the Minimal Risk model upsets the usual point of view because it aims to directly identify patients with chest pain and normal coronary arteries. Unfortunately, the only other external validation published up to the date of our search[57] cannot be considered here because it was based on a former version of Fordyce et al. 2017[34] that includes some computational errors[35].

With the exception of a few papers that discussed the classical DFM and DCS,[25,28,54] an in-depth study of the model performances with respect to operational cut-offs is mainly related to the CAD Consortium models and the Corus® CAD model. As far as the CAD Consortium models are concerned, clinical usefulness is assessed at cut-offs that vary from 5%-20%. A cut-off of 14.75 (15 in subsequent works) was identified for the Corus® CAD model in the main work,[46] a value that corresponds to a disease likelihood of 20% on a validation data set (positivity for index ≤ 15). Overall, sensitivity and specificity values are similar to those we derived for CADC-Clin: both these models show a higher balance between sensitivity and specificity than the guidelines and the DCS on the validation data sets. Finally, as suggested by the high values of sensitivity we derived in Table 3, the low AUC value of the uDFM obtained in Adamson[25] at the cut-off of 15% has been confirmed by Almeida,[28] Bittencourt[30] and, to a lesser extent, Zhang[54]; the corresponding AUC values are 0.52, 0.54 and 0.65 (for men) and 0.67 (for women). Analogous results come from the DCS's validations.[28,54]

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VALIDATED PTP MODELS: STRENGTHS AND WEAKNESSES

External validation is an indispensable tool for investigating the generalisability of a PTP model to populations that differ from the development population study. This process can utilise different approaches, from the computation of indexes to more complex procedures that aim at understanding how the original model should adapt to the new population. The papers included in this review mainly rely on AUC which only allows for a limited comparison among models. Different endpoint definitions and decision on whether or not to adopt a specific cut-off can yield different AUC values, as already highlighted. Moreover, only the whole receiver operating characteristic (ROC) curve will allow evaluation of the clinical usefulness of a test by showing the true positive and false positive fractions that will be obtained for any eventually chosen cut-off. As reported in our results, almost all the models provide a moderate discriminative ability (AUC from 0.70-0.80). Unfortunately, once the model is transferred into an operating scenario and the selection of a specific discriminatory cut-off is required, a clinical protection approach leads clinicians to prefer a very high sensitivity, which of course implies low specificity.[58,59] Only Corus® CAD,[48,50] CADC-Clin[47] and DCS[54] reach a moderate discriminative level at specified thresholds (0.71, 0.72 and 0.73, respectively). Notably, Corus® CAD recently lost Medicare coverage in the US.[60]

We believe that all the highlighted heterogeneity sources, and primarily those that refer to the outcome definition, prevent the direct derivation of any meta-analysis from the presented results.

Despite the fact that all the models are obtained by regression techniques, which allow the interpretation of the effect of the predictor on the outcome of interest, very few papers[32,37,40,47] address a complete validation procedure without rejecting a model after obtaining a poor preliminary performance on the new population by some test. Rather, a different model is developed, without any further in-depth analysis of the failure reason. Regardless of the quality of the new developed model, the lack of adequate consideration of in-depth validation procedures involves the loss of the information captured by the initial study and hinders a deep understanding of how effect size of relevant risk factors can change in a different geographical or setting framework.[24] For instance,

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deep validation procedures like miscalibration analysis allow questioning the effect of chest pain typicality in different data sets.[32,40,47] This finding is consistent with what was recently noted by Di Carli and Gupta:[61] angina remains a common presenting symptom in a high proportion of cardiac patients that do not show obstructive lesions in their coronary angiograms.

A central question is what clinical cardiologists are most interested in evaluating: CAD of any degree, high-risk plaques, stenosis of a certain anatomic/physiologic severity, stenosis that leads to ischaemia, stenosis that requires intervention or stenosis that must be fixed to reduce adverse outcomes. The answer determines which diagnostic pathway and test is the most appropriate[61,62] and also affects statistical analysis. A carefully defined outcome should be required to provide a reliable basis for the evaluation of the effect of any predictive variable.[63] When referring to validation specifically, the application of a model to predict an outcome different from the originally intended one raises some concerns and, eventually, should be explicitly noted. In data-driven models, the outcome definition in the population study also influences predictor selection. Thus, a small AUC value in the validation set does not necessarily indicate a lower performance of the original model on the new population. Instead, it suggests that the model may not be appropriate for the context.[57]

CONCLUSIONS

Several agencies and scientific organizations emphasise the need for increasing the knowledge on how the prediction of the disease can be modified according to the risk factors present in any specific study population or, possibly, in any particular patient. This would indeed improve the precision of the estimated clinical likelihood of CAD. However, the increasing availability of large data-sets, and the highly improved computational power seem to have directed large part of recent researches towards model development rather than model validation.[17] First of all, our review makes an important selection among the many developed models by mainly considering those externally validated. Then, it provides insights into the effects of traditional and emerging risk factors, biomarkers, and comorbidities on the PTP of obstructive CAD. Finally, our findings lead to the following important recommendations. To achieve a more robust exploitation of PTP models in

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decision-making processes, significant endpoints should be more clearly stated and consistently measured both in the derivation and validation phases. Furthermore, more comprehensive validation analyses should be adopted to understand model weaknesses. Finally, increased efforts are still needed to thresholds validation and to analyse the effect of PTP on clinical management.

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Conflict of interests

The authors declare that there is no conflict of interest.

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Authors' contributions

AB, CGL, PM, GP and SS provided substantial contribution to the conception of the work.

CGL and PM performed the literature search and retrieved selected publications.

All the authors contributed to the extraction and analysis of data.

AB, CGL, PM and MRT assessed the quality of included studies.

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All the authors contributed to draft the work. AB, CGL and PM revised it critically.

All the authors approved the version to be published and are accountable for all aspects of the work. CGL is responsible for the overall content as guarantor.

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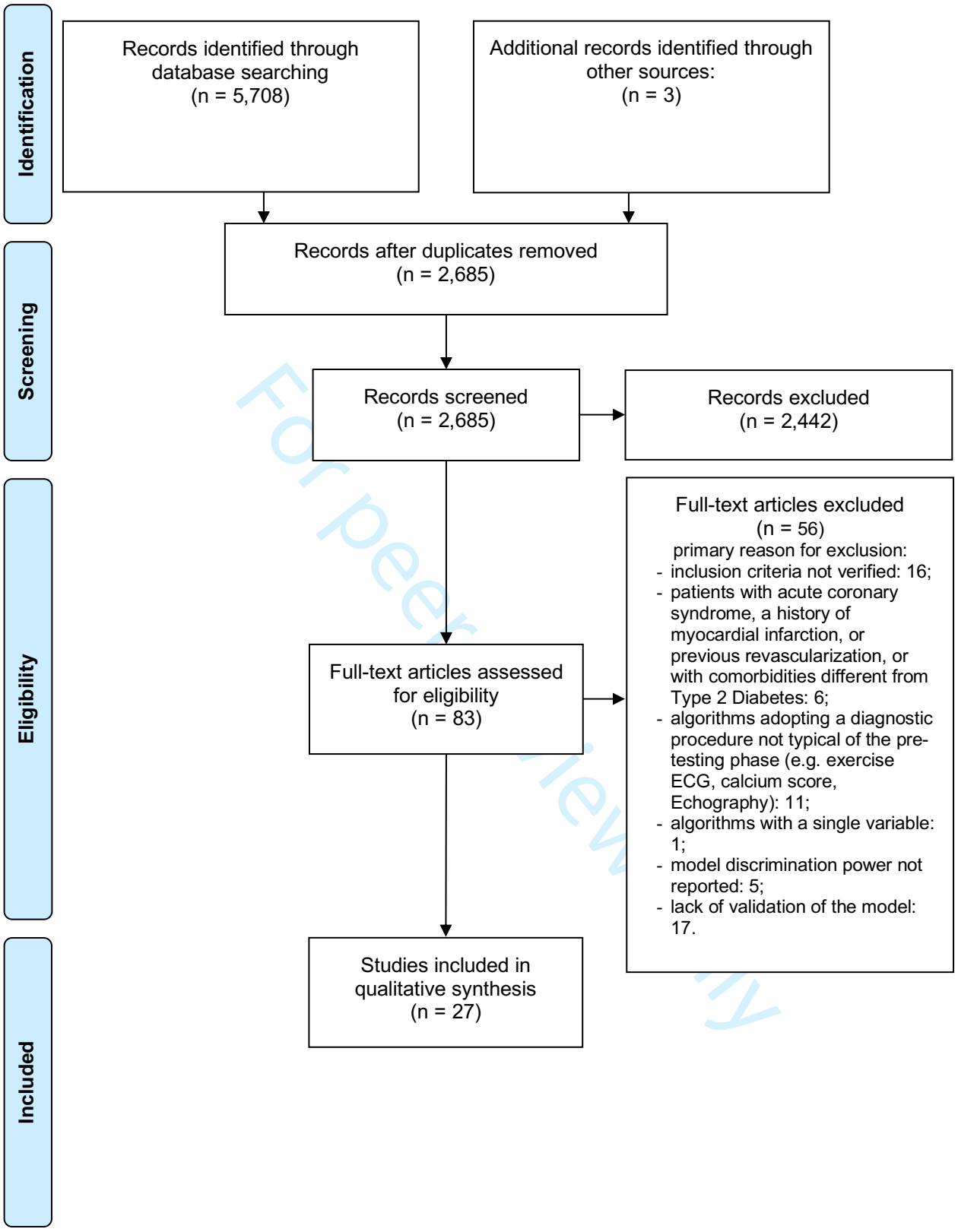
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Supplementary Material

Additional FILE 1 - Search strategy – It is the full search string adopted in OVID.

Additional File 2 – Study design and Eligibility Criteria of main studies – It provides details on the main studies cited in Table 1.

Additional File 3 – Proportion of studies with low, high or unclear risk of bias – It is a summary of the quality assessment according to QUADAS2.

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Additional FILE 1 - Search strategy

▼ **Search History** (28)

# ▲	Searches	Results
1	▶ Angina Pectoris/ or Angina Pectoris.af. or Angina, Stable/ or (Stable Angina* or Chronic Angina*).af.	98847
2	▶ Chest Pain/ or Chest Pain*.af.	72609
3	▶ Coronary Heart Disease/ or (CHD or Coronary Disease* or Coronary Heart Disease*).af.	412639
4	▶ Coronary Artery Disease/ or (CAD or Coronary Artery Disease* or Coronary Arteriosclerosis or Coronary Atheroscleros*).af.	288034
5	▶ Coronary Stenosis/ or (Coronary Stenos* or Artery Stenos*).af.	64518
6	▶ or/1-5	770062
7	▶ stratification score*.af.	492
8	▶ Likelihood Functions/ or Likelihood Function*.af. or (likelihood adj5 disease).af. or CAD likelihood.af. or "predict* CAD".af.	46652
9	▶ forecasting/ or (pre-test probabilit* or PTP or predictive model* or prediction or forecast*).af.	664543
10	▶ (probability adj5 disease).af.	8725
11	▶ or/7-10	716518
12	▶ Coronary Angiography/ or Angiograph*.af.	574196
13	▶ Angiocardiography/ or Angiocardiograph*.af.	21132
14	▶ Cardiac Catheterization/ or (Cardiac Catheterization* or Heart Catheterization*).af.	122622
15	▶ Computed Tomography Angiography/ or (Coronary Computed Tomography Angiograph* or CCTA).af.	21751
16	▶ or/12-15	677008
17	▶ 6 and 11 and 16	6069
18	▶ (ANIMALS not HUMANS).sh.	7244023
19	▶ 17 not 18	5985
20	▶ limit 19 to english language or limit 19 to italian	5708
21	▶ remove duplicates from 20	2682

Additional File 2 – Study design and Eligibility Criteria of main studies

Acronym	**COMPASS**
Name	Coronary Obstruction Detection by Molecular Personalized Gene Expression (Corus CAD or ASGES)
ClinicalTrials.gov Identifier	NCT01117506
Study design	Observational prospective study. The study enrolled a patient population that presented with stable chest pain syndrome or anginal equivalent and referred for stress myocardial profusion imaging
Study Center(s)	Multicenter trial: US (7 centers)
Inclusion Criteria:	• Ages 45-90 for women; 35-90 for men. • Stable chest pain syndrome (typical or atypical) or anginal equivalent in the judgment of the investigator (e.g., pain in the neck, jaw, arm or shoulder or dyspnea possibly due to cardiac ischemia). • Referred for a stress test using Myocardial Perfusion Imaging. • The patient has signed the appropriate Institutional Review Board approved Informed Consent Form
Exclusion Criteria	• History of known MI or significant CAD. • Current MI or acute coronary syndrome. • Current New York Heart Association (NYHA) class III or IV congestive heart failure symptoms. • Severe regurgitant or stenotic cardiac valvular lesion. • Severe left ventricular systolic dysfunction (LVEF ≤ 35 % documented in the last year); if no assessment was performed or documented in the year preceding enrolment, presume normal LVEF.

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- Active systemic infection (diagnosed by a combination of clinical symptoms and laboratory testing, including but not limited to fever, leukocytosis, positive blood cultures, pneumonia, urinary tract infection, or abscess in the preceding 2 months) or chronic infection (e.g., HIV, Hepatitis B or C, Tuberculosis).
- Protocol-specified rheumatologic, autoimmune or hematologic conditions (e.g., rheumatoid arthritis, systemic lupus erythematosis, polymyalgia rheumatica, or systemic sarcoidosis).
- Known or suspected diabetes mellitus or documented Hemoglobin A1c (HbA1c) ≥ 6.5; presume normal HbA1c if none documented.
- Total WBC ≥ 11,000 cells/ul and platelet count ≤ 75,000 cells/ul from a CBC with differential drawn within 7 days prior to enrollment [WBC ≥ 11,000 cells/ul and platelet count ≤ 75,000 cells/ul from a CBC drawn > 7 days prior need to be re-drawn at enrollment].
- Recipient of any organ transplant.
- Immunosuppressive or immunomodulatory therapy including any dose of systemic corticosteroids in the preceding 2 months.
- Chemotherapy in the preceding year.
- Major surgery in the preceding 2 months.
- Blood or blood product transfusion in the preceding 2 months.
- Subjects for whom all forms (stress or pharmacologic) of MPI are contraindicated.
- Subjects for whom invasive coronary angiography or coronary CT angiography is contraindicated, including IV beta-blocker.
- Subjects who planned to decline research CCTA or invasive coronary

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	angiography, regardless of MPI result. • Subjects with history of atrial fibrillation/flutter or frequent irregular or rapid heart rhythms. • Known history of renal insufficiency (serum creatinine ≥ 2.0 mg/dL), or severe allergy to iodinated contrast.
Acronym	**CONFIRM**
Name	COroNary CT Angiography Evaluation For Clinical Outcomes: An InteRnational Multicenter Registry
ClinicalTrials.gov Identifier	NCT01443637
Study design	Observational prospective study. Patients included in the CONFIRM Registry are those that have previously undergone clinically-indicated CCTA as part of their standard of care.
Study Center(s)	Multicenter trial: North America, Europe and Asia
Inclusion Criteria:	• Age > 18 years • Evaluation by CCTA with 64-detector rows or greater for CAD evaluation as part of standard of care • Interpretable CCTA • Prospective data collection for CAD risk factors.
Exclusion Criteria	• No explicit patient exclusion criteria are defined.
Acronym	**PREDICT**
Name	Personalized Risk Evaluation and Diagnosis (Using Corus CAD or ASGES) in

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	the Coronary Tree
ClinicalTrials.gov Identifier	NCT00500617
Study design	Observational prospective study. The study enrolled patients undergoing clinically indicated invasive coronary artery angiogram or CT angiogram
Study Center(s)	Multicenter trial: US
Inclusion Criteria:	• Age 21 to 99 Years • Referral for a coronary angiogram (either invasive X-ray angiography or coronary CTA) • Any one of the following clinical syndromes: ○ chest pain syndrome, stable angina, or anginal equivalent suggesting myocardial ischemia ○ low-risk unstable angin, or ○ asymptomatic individuals with a high probability of CAD.
Exclusion Criteria	• History of myocardial infarction or known CAD • Current Myocardial Infarct (MI), acute coronary syndrome with high-risk features or unstable angina with high-risk features • New York Heart Association (NYHA) class III or IV congestive • Inability to give informed congestive heart failures • Severe left ventricular systolic dysfunction (LVEF<35%) • Severe regurgitant or stenotic cardiac valve lesion • Active or chronic systemic infection • Rheumatologic, autoimmune or hematologic conditions • Any organ transplant • Immunosuppressive therapy

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	• Chemotherapy in the preceding year • Major blood or blood product transfusion in the preceding 2 months
Acronym	**PROMISE**
Name	PROspective Multicenter Imaging Study for Evaluation of Chest Pain
ClinicalTrials.gov Identifier	NCT01174550
Study design	Interventional (Clinical Trial) randomized study
Study Center(s)	Multicenter trial: North America
Inclusion Criteria:	• New or worsening chest pain suspicious for clinically significant coronary artery disease (CAD) • no prior evaluation for this episode of symptoms • planned non-invasive testing for diagnosis • men age ≥55 years • men age ≥45 years with increased probability of coronary artery disease (CAD) due to either (A. Diabetes Mellitus (DM) requiring medical treatment OR Peripheral Arterial Disease (PAD) defined as documented >50% peripheral arterial stenosis treated medically or invasively OR cerebrovascular disease (stroke, documented > 50% carotid stenosis treated medically or invasively) OR B. At least one of the following cardiovascular risk factors: 1-Ongoing tobacco use, 2-Hypertension, 3-Abnormal ankle brachial index (ABI) defined as less than <0.9, 4-Dyslipidemia • women age ≥65 years

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	• women age ≥50 years with increased probability of coronary artery disease (CAD) due to either (A. Diabetes Mellitus (DM) requiring medical treatment OR Peripheral Arterial Disease (PAD) defined as documented >50% peripheral arterial stenosis treated medically or invasively OR cerebrovascular disease (stroke, documented > 50% carotid stenosis treated medically or invasively) OR B. At least one of the following cardiovascular risk factors: 1-Ongoing tobacco use, 2-Hypertension, 3-Abnormal ankle brachial index (ABI) defined as less than <0.9, 4-Dyslipidemia • Serum creatinine ≤ 1.5 mg/dL within the past 90 days • Negative urine/serum pregnancy test for female subjects of child-bearing potential
Exclusion Criteria	• Diagnosed or suspected acute coronary syndrome (ACS) requiring hospitalization or urgent or emergent testing; Elevated troponin or creatine kinase-myocardial band (CK-MB) • Hemodynamically or clinically unstable condition systolic blood pressure (BP) < 90 mmHg, atrial or ventricular arrhythmias, or persistent resting chest pain felt to be ischemic despite adequate therapy) • Known coronary artery disease (CAD) with prior Myocardial infarction (MI), percutaneous coronary intervention (PCI), coronary artery bypass graft (CABG) or any angiographic evidence of coronary artery disease (CAD) ≥50% lesion in a major epicardial vessel • Any invasive coronary angiography or non-invasive anatomic or functional cardiovascular (CV) test for detection of coronary artery

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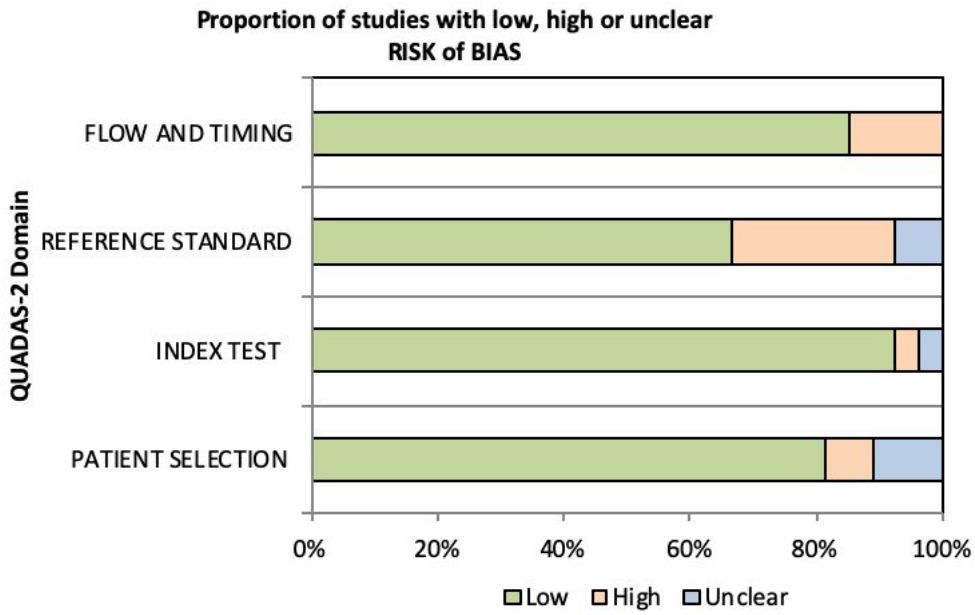
	disease (CAD), including coronary tomographic angiography (CTA) and exercise electrocardiogram (ECG), within the previous twelve (12) months • Known significant congenital, valvular (> moderate) or cardiomyopathic process (hypertrophic cardiomyopathy or reduced systolic left ventricular function (LVEF) ≤ 40%)) which could explain cardiac symptoms • Contraindication to undergoing a coronary tomographic angiography (CTA), including but not limited to: a. Allergy to iodinated contrast agent, b. Unable to receive beta blockers unless heart rate < 65 beats per minute, c. Pregnancy • Life expectancy < 2 years • Unable to provide written informed consent or participate in long-term follow-up
Acronym	**SCOT-HEART**
Name	Scottish COmputed Tomography of the HEART Trial
ClinicalTrials.gov Identifier	NCT01149590
Study design	Interventional (Clinical Trial) randomized study
Study Center(s)	Multicenter trial: Scotland (UK)
Inclusion Criteria:	• 18 and ≤75 years of age • Attendance at the Rapid Access Chest Pain Clinic
Exclusion Criteria	• Inability or unwilling to undergo computed tomography scanning, such as exceeding weight tolerance of scanner

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- Severe renal failure (serum creatinine >200 µmol/L or estimated glomerular filtration rate <30 mL/min)
- Previous recruitment to the trial
- Major allergy to iodinated contrast agent
- Unable to give informed consent
- Known pregnancy
- Acute coronary syndrome within 3 months

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Additional File 3 – Proportion of studies with low, high or unclear risk of bias



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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Additional file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	N.A.

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Discrimination capability of pre-test probability of stable coronary artery disease: a systematic review and meta-analyses suggesting how to improve validation procedures

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Discrimination capability of pre-test probability of stable coronary artery disease: a systematic review and meta-analyses suggesting how to improve validation procedures

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Word Count: 3965

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ABSTRACT

Objective: Externally validated pre-test probability models for risk-stratification of subjects with chest pain and suspect stable coronary artery disease (CAD) determined through invasive coronary angiography or coronary computed tomography angiography, are analysed to characterize the best validation procedures in terms of discriminatory ability, predictive variables and methods completeness.

Design: Systematic review and meta-analysis.

Data sources: Global Health (Ovid), Healthstar (Ovid) and MEDLINE (Ovid) searched on 22 April 2020.

Eligibility criteria: We included studies validating pre-test models for the first line assessment of patients with chest pain and suspected stable CAD. Reasons of exclusion: acute coronary syndrome, unstable chest pain, a history of myocardial infarction or previous revascularisation; models referring to diagnostic procedures different from the usual practices of the first-line assessment; univariable models; lack of quantitative discrimination capability.

Methods: Eligibility screening and review were performed independently by all the authors. Disagreements were resolved by consensus among all the authors. The quality assessment of studies conforms to QUADAS-2. A random effects meta-analysis of area under the ROC curve values (AUC) for each validated model was performed.

Results: 27 studies were included for a total of 15 models. Beside age, sex, and symptom typicality, other risk factors are smoking, hypertension, diabetes mellitus and dyslipidaemia. Only one model considers genetic profile. AUC values range from 0.51-0.81. Significant heterogeneity ($p < 0.003$) was found in all but two cases ($p > 0.12$). Values of $I^2 > 90\%$ for most analyses and not significant meta-regression results undermined relevant interpretations. A detailed discussion of individual results was then carried out.

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Conclusions: We recommend a clearer statement of endpoints, their consistent measurement both in the derivation and validation phases, more comprehensive validation analyses and the enhancement of threshold validations to assess the effects of pre-test models on clinical management.

Trial registration number PROSPERO CRD42019139388

Strengths and limitation of this study

- This is the first meta-analysis summarising the most up-to-date data on the discrimination capability of pre-test probability models of stable CAD.
- The systematic review pays careful attention to the whole validation procedures.
- The majority of included studies were considered to be of high methodological quality.
- We considered pre-test models developed in cohorts of patients referred for an anatomical test.
- The meta-analyses has a low reliability due to the small number of included studies and the very high heterogeneity.

Keywords: Coronary Heart Disease; Cardiovascular imaging; Computed tomography; PUBLIC HEALTH

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INTRODUCTION

The leading cause of mortality and morbidity worldwide in 2019 were represented by Cardiovascular Diseases (CVDs) with 523 million prevalent cases and 18.6 million deaths.[1] Among these, Coronary Artery Disease (CAD) was reported in 197 million subjects and caused 9.14 million deaths. Stable CAD is typically caused by the buildup of plaques that limit blood flow and is characterized by reversible myocardial demand/supply mismatch usually inducible by exercise, emotion or other stress and commonly associated with transient chest pain (stable angina pectoris).[2,3]

Stable CAD diagnosis is supported by non-invasive functional and/or anatomical testing,[2,3] and invasive coronary angiography (ICA).[2] To limit the risk of inappropriate examinations and their consequences on patients' and healthcare professionals' safety, and economic sustainability of Healthcare Systems,[4–7] eligibility to diagnostic testing is established through models that provide a risk stratification of subjects based on a pre-test probability (PTP) of CAD. Since the introduction of the Diamond-Forrester model (DFM)[8] and the Duke Clinical Score (DCS)[9], several alternative PTP models have been developed in cohorts of patients referred for ICA or coronary computed tomography angiography (CCTA). Indeed, due to its very high sensitivity and negative predictive values, CCTA can substantially contributes to rule-out CAD.[10] The DFM and its more recent updates have been recommended in guidelines for stable symptomatic subjects.[3,11] Recent debates within scientific societies broach the question of the overestimation flaw of such models. The UK National Institute for Health and Care Excellence (NICE) has preferred no longer to resort to a probabilistic risk-stratification approach and adopt a simpler identification of anginal chest pain to decide for further testing.[12] The European Society of Cardiology (ESC) updated guideline determines PTPs from the stratified prevalence of CAD in a contemporary cohort, instead of recurring to a prediction model as in the past. These new estimated risks are noticeably lower compared to

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the previous ones and then underestimation of the disease prevalence can be obtained in different populations.[13] US Experts are debating on whether adopting the NICE diagnostic approach or keeping on using PTP.[14,15] To face the flaws on widely recognized PTP models highlighted by NICE and ESC, these organizations clearly underline the need for more information on the various risk factors acting as modifier of the PTP, especially in the low probability range,[11] and for the development and validation of new scores addressing outstanding uncertainties in the estimation of the PTP of CAD.[12]

This review provides several new contributions to the actual debate on how to ameliorate the PTP models developed for anatomically defined outcomes. It focuses on external validation mainly,[16] carries out a meta-analysis to identify the best results and characterizes the best procedures in terms of discriminatory ability, significant predictive variables and methods completeness. By highlighting some key issues that could be further improved in the development and validation phases, this work aims at stimulating more rigorous procedures for the comparison of different pre-test models.

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METHODS

This systematic review conforms to the PRISMA statement;[17] the protocol was registered in PROSPERO (CRD42019139388).[18]

Study inclusion and exclusion criteria

We identified studies that validated pre-test models intended for the first line assessment of patients with chest pain and a suspect of stable CAD. The disease was considered as a binary outcome determined through either ICA or CCTA. Reasons of exclusion were: (i) acute coronary syndrome, unstable chest pain, a history of myocardial infarction or previous revascularisation; (ii) models that included a diagnostic procedure that do not reflect the usual practices of the first-line assessment;[3,11] (iii) models based on a single predictive variable; (iv) lack of clearly stated discrimination capability. Unlike previous works,[19] external validation was primarily considered. We also included internal validation but limited it to *k*-fold cross-validation as a technique inspired by the same purposes of external validation. Moreover, papers referring to Machine learning-based PTP models have been excluded as considered in a recent review focusing on CAD diagnosis by ML with aims close to ours.[20] Only full papers were retained because other publications, e.g., letters to editors, conference proceedings, et cetera, are usually not assessed for study quality. Only articles published in English and Italian were considered.

Searches

The databases Global Health (Ovid), Healthstar (Ovid) and MEDLINE (Ovid) were systematically searched (CGL, PM) on 22 April 2020 using several keywords, including: angina pectoris, chest pain, coronary artery disease, coronary heart disease, coronary stenosis, stratification score, likelihood function, predictive model, pre-test probability, coronary angiography, cardiac catheterisation and computed tomography angiography. The same full

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electronic search strategy was applied to the three databases (no filter was used), and is reported in Additional File 1. Citation searches were also performed on reference lists of definitively included studies.

Study selection

Eligibility screening was performed independently by all the authors. Preliminary screening was performed using Abstrackr[21] based on title and abstract with each paper assessed by two randomly assigned reviewers out of the authors. Selected papers were assessed based on full text. Disagreements were resolved by consensus among all the authors.

Data extraction strategy

A data collection form was developed by three authors (AB, CGL, PM) and filled by reviewers independently. Each selected paper was assigned for data extraction to the statistician (AB) and two randomly selected reviewers. Correspondence with the authors of the included studies was initiated if necessary. The reviewers worked independently and in plenary session meetings. Disagreements were resolved by consensus among all the authors. AB, CGL and PM reviewed the final form for internal consistency.

Study quality assessment

The quality assessment of included studies conforms to QUADAS-2 and was performed by four reviewers (AB, CGL, PM, MRT).[22] Due to the previously described features (i)-(iv), we considered that the eligible works did not raise applicability concerns.

Data synthesis and statistical analysis

The discriminative performances of prediction models can be summarised using several methods and indices, and the area under receiver operating characteristic curve (AUC) or *c*-statistics is certainly the best-known and more suitable.[23] Then, it has been chosen as main index for the purposes of this review. Sensitivity and specificity also describe the discrimination capability of the model for a given cut-off and thus provide an indication of

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clinical usefulness. However, the bivariate nature of this index is not suitable for direct comparisons and then we resorted to the associated AUC.

For the purposes of generalisation of a PTP model to populations that differ from the development population study, the computation of performance indexes is not sufficient because a lower performance is usually expected.[16,24] Therefore, we also noted whether more extended validation procedures were performed in order to properly apply a model to new populations.

A random-effects meta-analysis of AUC values from validations of each identified model was performed using R Statistical Software (R Project for Statistical Computing, RRID:SCR_001905)[25] by meta[26] and auctestr[27] packages. Meta-regression was planned to explore the possible sources of unexplained heterogeneity by considering the following factors: (1) sample size, (2) prevalence, (3) anatomical test for outcome assessment.

Patient and public involvement

Patients and the public were not involved in this review.

Ethics approval statement

Our work of systematic review and meta-analyses of published literature doesn't require a Research Ethics Approval.

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RESULTS

Study selection

A total of 5,711 studies were identified (3 through reference lists of included studies) and 2,685 different abstracts were screened. Out of the 71 relevant full-texts assessed for eligibility, 27 were finally included (Figure 1).

Study characteristics

Table 1 summarises the selected studies in terms of model name, geographical location, population recruitment criteria. Sometimes the same model is referenced with different names across the papers, then Table 1 indicates the original name and the one we adopted here.

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Table 1 Characteristics of the studies on PTP for CAD

Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Adamson PD, 2018a[28]	DFM/CASS uDFM	1) Multicenter PROMISE trial, US & Canada 2) Multicenter SCOT-HEART trial, Scotland (UK)	See PROMISE. Randomised to receive CCTA as non-initial non-invasive test. See SCOT-HEART. Randomized to the CCTA intervention arm.	See PROMISE & SCOT-HEART. Known CAD
Adamson PD, 2018b[29]	uDFM (Baseline CADC model, in text) uDFM-cTn (Baseline CADC model with the addition of troponin, in text)	Odense University Hospital, Denmark	Clinical stable prospectively enrolled patients with suspected angina pectoris scheduled for either ICA or CCTA[30]	Suspected acute coronary syndrome. To avoid potential confounding effects on the biomarkers measured, patients with established atherosclerotic manifestations, including an abnormal 12-lead rest electrocardiogram, were excluded: known ischemic heart disease, prior ischemic stroke or transitory ischemic attack, known peripheral artery disease (n = 10), and p-creatinine >200 mmol/L. CCTA not performed or of poor technical quality, lack of informed consent, missing hs-cTnI measure or personal history.[30]

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Almeida J, 2016[31]	CADC-Clin (CAD Consortium 2, in text) DCS uDFM (CAD Consortium 1, in text)	Single center in Southwestern Europe	Patients with chest pain and suspected CAD referred to ICA	Patients with a history of CAD, acute coronary syndrome, or coronary revascularization
Baskaran L, 2018[32]	CADC-Clin CONFIRM score uDFM	Multicenter SCOT-HEART trial, Scotland (UK)	See SCOT-HEART. Randomized to the CCTA intervention arm and with information on all variables needed for the analysis.	See SCOT-HEART. Known CAD
Bittencourt MS, 2016[33]	CADC-Basic CADC-Clin uDFM (Diamond and Forrester score, in text)	Massachusetts General Hospital; Brigham and Women's Hospital (Massachusetts, USA)	Subjects ≥ 18 years who underwent CCTA for suspect of CAD	Patients who were missing any of the clinical information needed to calculate the pre-test probability, who had nondiagnostic CCTA images, who had incomplete follow-up information; with congenital heart disease, heart transplantation, or prior CAD, defined as prior percutaneous coronary interventions, coronary artery bypass graft surgery, or MI

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Daniels SE, 2014[34]	Corus® CAD (Gene Expression score – GES, in text)	Multicenter PREDICT trial US	See PREDICT	See PREDICT. Diabetic patients
Edlinger M, 2017[35]	CADC-Clin	University Clinic of Cardiology at Innsbruck (Austria)	Patients were 18 years of age or older with chest pain or symptoms suggestive of CAD (predominantly dyspnoea) and/or non-invasive evidence of CAD referred for elective ICA.	1) an elective ICA before or after heart transplantation, 2) an elective ICA prior to solid organ transplantation, 3) an elective ICA before heart valve repair or replacement, or with valvular heart disease as leading clinical diagnosis, 4) an isolated right heart catheterisation, 5) an electrophysiological procedure (pace-maker implantation or catheter ablation) as leading clinical indication, 6) an elective ICA because of a known or suspected congenital heart disease as leading clinical diagnosis (e.g., atrial septal defect, ventricular septal defect or patent foramen ovale), or 7) when referred for other reasons (like myocardial biopsy, aortic aneurysms, myxoma, endocarditis or prior failed angiography). History of myocardial infarction.

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Ferreira AM, 2016[36]	uDFM (Modified DF, in text) CADC-Clin (CAD consortium 2, in text) CONFIRM score	Unspecified, Portugal	Patients undergoing CCTA for the evaluation of CAD	Age <30 years; known CAD; suspected acute coronary syndrome; preoperative assessment; known left ventricular systolic dysfunction; asymptomatic patients (typically referred after a positive screening exercise test); symptoms other than chest pain. Patients with suspected CAD who were scheduled to undergo CCTA but had the procedure halted due to a high coronary artery calcium (CAC) Agatston score. A threshold of 400 was used as a general guideline for withholding CCTA in these circumstances, but the decision was ultimately left to the performing physician, taking into consideration the clinical context and the distribution of calcium in the coronary tree.
Fordyce CB, 2017[37]	PROMISE minimal risk model (The originally published version has been subsequently corrected	Multicenter PROMISE trial, US & Canada	See PROMISE. Patients assigned to anatomic testing	See PROMISE

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
	online, see Fordyce CB, 2018[38])			
Fujimoto S, 2014[39]	DCS K-score	Multicenter, Japan	Suspected CAD	Patients with known CAD, showing poor image quality and patients with un-assessable segments due to severe calcification
Genders TSS, 2011[40]	DFM	14 European centers	Patients aged 30-69 with stable chest pain (typical, atypical, or non-specific chest pain) and if ICA performed.	Patients meeting the following criteria: (i) acute coronary syndrome or unstable chest pain, (ii) history of myocardial infarction or previous revascularization (percutaneous coronary intervention or coronary artery bypass graft surgery), and (iii) no informed consent.
	uDFM	Erasmus Medical Center, Rotterdam, the Netherlands[41]	Patients with stable chest pain and no history of CAD[41]	Not undergoing CCTA or ICA
Genders TSS, 2012[42]	DCS	Multicenter EU and US	Stable chest pain, referred for catheter based or CT based coronary angiography	Acute coronary syndrome, unstable chest pain, history of myocardial infarction or previous revascularization or no informed consent.

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
	CADC-Basic CADC-Clin	Multicenter EU and US	Stable chest pain, referred for catheter based or CT based coronary angiography	Acute coronary syndrome, unstable chest pain, history of myocardial infarction or previous revascularization or no informed consent.
Genders TSS, 2018[43]	CADC-Basic CADC-Clin	Multicenter PROMISE trial, US & Canada	See PROMISE Trial for the main criteria. Patients assigned to anatomic testing	See PROMISE Trial for the main criteria
Jensen JM, 2012[44]	CORSCORE DCS DFM Morise score uDFM	Lillebælt Hospital Vejle, Denmark	Patients with chest pain indicative of CAD referred for ICA	Unstable angina or previous coronary intervention
Min JK, 2015[45]	CONFIRM score (Integer-based risk model, in text)	United States, Canada, South Korea and Austria (4 out of 5 sites of the Phase II of	Patients ≥18 years old referred to CCTA for suspected stable CAD (CONFIRM trial[46])	Patients with prior coronary revascularization or MI, asymptomatic, missing data

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
		CONFIRM trial[46])		
Pickett CA, 2013[47]	DFM/CASS Morise score	Walter Reed Army Medical Center, Washington USA	Patients referred for CCTA	Known CAD
Rademaker AA, 2014[48]	DCS DFM Morise score (New score, in text) uDFM	VU University Medical Center, Amsterdam, The Netherlands	Symptomatic women undergoing evaluation for CAD and referred for CCTA	Prior history of CAD (percutaneous coronary intervention, coronary artery bypass graft surgery, or previous myocardial infarction), or absolute or relative contraindications for CCTA such as (i) significant severe arrhythmia; (ii) pregnancy; (iii) renal insufficiency (glomerular filtration rate<45 ml/min); (iv) known allergy to iodinated contrast material.
Rosenberg S, 2010[49]	Corus® CAD (Gene expression test, in text) Expanded clinical model score	Multicenter PREDICT trial US	See PREDICT	See PREDICT. Diabetes

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
	DFM/CASS			
Teressa G, 2018[50]	CADC-Basic CADC-Clin	1 center in US	>18 years old evaluated in the Emergency Department of a major academic tertiary university hospital for chest pain, using CCTA as a primary diagnostic modality	Known CAD, defined as history of acute myocardial infarction, percutaneous intervention, coronary artery bypass graft, or evidence of CAD by either anatomical (CCTA or cardiac catheterization) or functional tests (positive stress test). Hemodynamically or clinically unstable patients, patients with ST segment changes or positive cardiac troponin (>0.04ng/ml), impaired renal function (eGFR<50ml/min/1.73m^2), tachycardia, or contraindication to nitroglycerin or iodinated contrast. Inadequate documentation on Chest pain characteristics, repeat CCTAs, unavailable calcium score and non-diagnostic exam.
Thomas GS 2013[51]	Corus® CAD (GES, in text) DFM Morise score	Multicenter COMPASS trial, US	See COMPASS	See COMPASS

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Voora D, 2017[52]	Corus® CAD	Multicenter PROMISE trial, US & Canada	See PROMISE. Patients assigned to anatomic testing	See PROMISE. Diabetes. RNA sample not passing quality control.
Voros S, 2014[53]	Corus® CAD (GES, in text) DFM	Multicenter PREDICT US and COMPASS US trials	See PREDICT and COMPASS.	See PREDICT and COMPASS. Diabetes excluded from PREDICT cohort.
Wang M 2018[54]	CONFIRM score	Not specified, China	Patients who underwent CCTA for stable chest pain and with 0 or 1 risk factors among smoking, hypertension, diabetes and hyperlipidemia	Acute coronary syndrome, previous CAD or coronary revascularization, un-assessable segments due to motion artifact, atrial fibrillation, aortic disease, New York Heart Association class III or IV heart failure, age > 90 years old, pacemaker leads or missing data
Winther S, 2019[55]	uDFM CADC-Basic CADC-Clin	Multi-center Dan-NICAD trial, Denmark	Patients without known CAD referred to CCTA due to a history of symptoms suggestive of CAD	Age <40; previous coronary revascularization or MI; unstable angina pectoris; estimated glomerular filtration rate <40mL/min; pregnancy; and contraindication for iodine-containing contrast medium, magnetic resonance imaging, or adenosine (severe asthma, advanced atrioventricular block, or critical aortic stenosis).

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Yang Y, 2015[56]	High Risk Anatomy (HRA) score	Multicenter CONFIRM trial,[46] North America, Europe and Asia University of Ottawa Heart Institute Cardiac CT registry	Patients ≥18 years old referred to CCTA for suspected stable CAD (CONFIRM trial)[46]	Documented CAD, history of MI, coronary revascularization, cardiac transplantation, congenital heart disease
	uDFM	Multicenter CONFIRM trial,[46] North America, Europe and Asia	Patients ≥18 years old referred to CCTA for suspected stable CAD (CONFIRM trial)[46]	Documented CAD, history of MI, coronary revascularization, cardiac transplantation, congenital heart disease

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Study	Models / scores	Study Centers	Population	
			Inclusion criteria	**Exclusion criteria**
Zhang Y, 2019[57]	DCS uDFM	Tianjin Chest Hospital, Tianjin, China	Patients with stable chest pain and referred for CCTA	Acute coronary syndrome, previous CAD or coronary revascularization (percutaneous coronary intervention or coronary artery bypass grafting), impaired renal function (serum creatinine > 120 μmol/l), New York Heart Association class III or IV heart failure, atrial fibrillation, aortic disease, age more than 90 years, or patients with un-assessable segments because of artefact
Zhou J, 2017[58]	CADC-Clin (Genders clinical model, in text) DCS uDFM	Not specified, China	Patients who underwent CCTA for stable chest pain	Acute coronary syndromes, previous CAD or coronary revascularization (percutaneous coronary intervention or coronary artery bypass grafting), patients with un-assessable segments due to motion artefact, atrial fibrillation, aortic disease, New York Heart Association class III or IV heart failure, age > 90 years, presence of pacemaker leads or missing data.

The trials COMPASS, CONFIRM, PREDICT, PROMISE and SCOT-HEART were considered in several studies, and thus their main characteristics are fully reported in Additional File 2

Legend

CADC-Basic = CAD Consortium Basic model

CADC-Clin = CAD Consortium Clinical model

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CASS = Coronary Artery Surgery Study

DCS = Duke Clinical Score

DFM = Diamond & Forrester (DF) model

eGRF = estimated glomerular filtration rate

NA = Not Available

QCA = Quantitative Coronary Angiography

uDFM = Updated Diamond-Forrester model

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Studies are mainly conducted in North America [33,34,37,43,47,49–53] or Europe[29,31,32,35,36,40,44,48,55].

The Updated Diamond-Forrester model (uDFM),[28,29,31–33,36,40,44,48,55–58] and the CAD consortium clinical model (CADC-Clin)[31–33,35,36,42,43,50,55,58] are the most assessed models.

The quality of included studies is generally high due to the specific review question and adopted eligible criteria. Nevertheless, risk of bias arises from a few specific issues. A few validation studies[34,40,42,49,53] do not declare that they enrolled only consecutive or random samples of patients. With respect to the index test, only one work adopted an optimal discriminating threshold in addition to pre-specified ones.[53] Application of CCTA as a reference test yields a risk of bias in many studies[28,37,40,42,43,45,48,52,56] that do not report measures against misclassification of the test results. Finally, in four works[29,42,43,51] patients did not receive the same reference test for the diagnosis of stable CAD. A graphical summary of the risk of bias is reported in Additional File 3.

Predictive variables

As shown in Table 2, the identified models can be classified into two broad classes: basic models, including the DFM (based on age, sex and chest pain) and its updates, and clinical models, including the DCS and the models that extend the DFM by adding a few, mainly traditional,[59] risk factors. Within this quite classical framework, the Corus® CAD model is distinguished by relating CAD in nondiabetic patients to the expression levels of a set of genes.

All the models were derived by logistic regression. Exceptions are: DFM, derived by a conditional probability analysis in the late 1970s; Corus® CAD, obtained through Ridge regression; CONFIRM score, developed to predict adverse clinical events by fitting a Cox proportional hazards model and subsequently validated for diagnosis of CAD.

Cross-validation[42] and split-sample[37,49] have been used in a few cases only.

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Table 2 PTP models' variables

Macro categories	Predicting variables	Model/Score															
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
		[33,42,43,50,55]	[31–33,35,36,42,43,50,55,58]	[32,36,45,54]	[44]	[34,49,51–53]	[31,39,42,44,48,57,58]	[40,44,45,48,51,53]	[28,47,49]	[49]	[39]	[56]	[44,47,48,51]	[37]	[28,29,31–33,36,40,44,48,55–58]	[29]
Demography	Age	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	Sex	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	Race									√				√		
Medical history	Diabetes mellitus		√	√			√				√	√	√	√		
	Hypertension		√	√	√					√	√	√	√	√		

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Macro categories	Predicting variables	Model/Score															
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
	Previous MI				√		√									
	Cerebral Infarction										√					
	Peripheral vascular disease											√				
Clinical presentation/ physical examination	Chest pain	√	√	√	√		√	√	√	√	√	√	√		√	√
	Abnormal ECG						√									
	Obesity												√			
	Smoking		√	√	√		√				√	√	√	√		

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Macro categories	Predicting variables	Model/Score														
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
	Family history of CAD			√								√	√	√		
	Other (specify)				Medically treated hypercho l.					Medically treated hypercho l.				Symptoms related to physical or mental stress		
Bio-Chemistry	HDL cholesterol													√		
	Dyslipidaemia		√				√			√	√	√	√	√		

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Macro categories	Predicting variables	Model/Score														
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
	Oestrogen status												√			
	Gene expression					√										
	Troponin															√
Others										Aspirin, anti-platlet, ACE inhibitor use, systolic						

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Macro categories	Predicting variables	Model/Score														
		CADC-basic	**CADC-Clin**	**CONFIRM score**	**CORSCORE**	**Corus® CAD**	**DCS**	**DFM**	**DFM/CASS**	**Expanded clinical model score**	**K-score**	**HRA score**	**Morise score**	**PROMISE Minimal Risk model**	**uDFM**	**uDFM-cTn**
										blood pressure						
Derivation method		Log	Log	Cox proportional hazards models	Log	Score derived by a Ridge regression	Log	Conditional probability analysis*	Log	Log	Log	Score derived by a multivariable Log	Score derived by a Log	Log	Log	Log

* In Genders, 2011,[40] to unravel the implicit coefficients of the predictors in this model, the authors performed a weighted linear regression on the log odds of the DF predictions per subgroup

Legend

DCS = Duke Clinical Score

DFM = Diamond & Forrester model

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Log = Logistic regression

QCA = Quantitative Coronary Angiography

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Predictors were classified into four macro-areas: demography, medical history, clinical presentation/physical examination and biochemistry. The demographic macro-area is present in all models with the variables age and sex, while race is only included in the Expanded clinical model and PROMISE Minimal Risk model. The most used variables in the medical history macro-area are diabetes mellitus and hypertension. The clinical presentation/physical examination macro-area is present in all but the Corus® CAD models. Only the Corus® CAD and PROMISE Minimal Risk models do not include chest pain. The most used variable in the biochemistry macro-area is dyslipidaemia. The other risk factors are model-specific: gene expression (Corus® CAD), oestrogen status (Morise score), high-density lipoprotein cholesterol (PROMISE Minimal Risk model) and the high-sensitivity cardiac troponin (uDFM-cTn).

Discrimination capability

All the papers presented ROC curves and/or AUC values. In Adamson and colleagues[28] fixed thresholds only were analysed and the *c*-statistics associated with sensitivity and specificity reported. Table 3 reports the AUC values and their 95% confidence intervals, while the summary of the meta-analyses conducted for the models with more than one validation are shown in Figure 2, where models with a single validation are also considered for the sake of completeness. To carry out meta-analyses as complete as possible, the missing information about the standard error of estimated AUC values was filled in by the "se_auc" command of the auctestr package. Then, the (gaussian) 95% confidence intervals are reported in Table 3. This computation only requires to know the study sample size and the prevalence, and is as better as the size of the study is larger. For a small sample size, the computed standard error is generally larger than the exact one and then confidence intervals are more conservative. For only two papers, the conditions for inclusion in the meta-analyses are not met.[34,37]

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Table 3 AUC values of PTP models

Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
CADC-Basic	Bittencourt MS, 2016[33]	At least 1 segment (with a >2 mm diameter) with a lesion with ≥50% diameter stenosis	CCTA	2,274	22	0.7517 (0.729 - 0.775)
	Genders TSS, 2012[42]	≥1 diameter stenosis of ≥50% in ≥1 vessel	CCTA, ICA	min: 471 max: 1,241	NA	Mean: 0.77
	Genders TSS, 2018[43]	≥1 diameter stenosis of ≥50% in ≥1 vessel (≥2.0 mm diameter) by ICA. Patients with a completely normal CCTA (0% stenosis and coronary artery calcium score of 0) are considered as free of obstructive CAD on ICA.	CCTA, ICA	3,468	23	0.69 (0.67 - 0.72)
	Teressa G, 2018[50]	1 vessel with stenosis of 50%	CCTA	1,981	10.4	0.77 (*0.731 - 0.809*)
	Winther S, 2019[55]	Coronary diameter stenosis reduction ≥50% in all segments with a reference vessel diameter >2mm	CCTA	1,653	23.7	0.66 (0.63–0.69)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
CADC-Clin	Almeida J, 2016[31]	Stenosis of >50% in at least one major epicardial vessel	ICA	2,234	58.5	0.683 (0,661 - 0.706)
	Baskaran L, 2018[32]	A stenosis causing ≥50% diameter stenosis	CCTA	1,738	37.7	0.790 (0.768 - 0.811)
	Bittencourt MS, 2016[33]	At least 1 segment (with a >2 mm diameter) with a lesion with ≥50% diameter stenosis	CCTA	2,274	22	0.791 (0.770 - 0.812)
	Edlinger M, 2017[35]	Stenosis ≥50% diameter in at least one of the main coronary arteries	ICA	4,888	44	0.69 (0.67 - 0.70)
	Ferreira AM, 2016[36]	Coronary diameter stenosis ≥50%	CCTA	1,069	13.8	0.73 (0.71 - 0.76)
	Genders TSS, 2012[42]	≥1 diameter stenosis of ≥50% in ≥1 vessel	CCTA, ICA	min: 471 mean: NA max: 1,241	NA	0.78 0.79 0.81
	Genders TSS, 2018[43]	≥1 diameter stenosis of ≥50% in ≥1 vessel (≥2.0 mm diameter) by ICA. Patients with a completely normal CCTA (0% stenosis and coronary artery calcium score of 0) are considered as free of obstructive CAD on ICA.	CCTA, ICA	3,468	23	0.72 (0.69 - 0.74)
	Teressa G,	1 vessel with stenosis of 50%	CCTA	1,981	10.4	0.80

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
	2018[50]					(0.763 - 0.837)
	Winther S, 2019[55]	Coronary diameter stenosis reduction ≥50% in all segments with a reference vessel diameter >2mm	CCTA	1,653	23.7	0.69 (0.66–0.72)
	Zhou J, 2017[58]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	5,743	32.6	0.774 (0.761 - 0.788)
CONFIRM score	Baskaran L, 2018[32]	A stenosis causing ≥50% diameter stenosis	CCTA	1,738	37.7	0.749 (0.726 - 0.771)
	Ferreira AM, 2016[36]	Coronary diameter stenosis ≥50%	CCTA	1,069	13.8	0.71 (0.66 - 0.75)
	Min JK, 2015[45]	≥50% luminal diameter stenosis in any coronary artery ≥1.5 mm in diameter	CCTA	2,132	NA	0.76 (0.746 - 0.771)
	Wang M 2018[54]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	0 Risk Factors (RF): 1,201	30.2	0.756 (0.731 - 0.781)
				1 RF: 2,415	27.1	0.762 (0.742 - 0.783)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
CORSCORE	Jensen JM, 2012[44]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.727 (*0.684-0.770*)
Corus® CAD	Daniels SE, 2014[34]	At least one lesion in a major coronary artery (≥1.5 mm lumen diameter) ≥70% diameter stenosis by clinical read or ≥50% diameter stenosis by invasive QCA	ICA	Several subsets from a total of 1,502	NA	min: 0.64 max: 0.72
	Rosenberg S, 2010[49]	≥1 atherosclerotic plaque in a major coronary artery (≥1.5 mm lumen diameter) causing ≥50% luminal diameter stenosis by QCA	ICA	526	36.5	0.70 (0.68 - 0.72)
	Thomas GS 2013[51]	≥1 diameter stenosis ≥50% in a major vessel on ICA by QCA (≥1.5 mm) or CCTA (≥2.0 mm)	CCTA, ICA	431	14.6	0.79 (0.72 - 0.84)
	Voora D, 2017[52]	≥70% stenosis in major coronary artery or ≥50% left main stenosis	CCTA	1,137	10.1	0.625 (0.573 - 0.678)
	Voros S, 2014[53]	Outcome 50: ≥ 50% maximum diameter stenosis	CCTA	610	14	0.75 (0.70 - 0.80)
		Outcome 70: ≥ 70% maximum diameter stenosis	CCTA		NA	0.75 (0.67 - 0.83)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
DCS	Almeida J, 2016[31]	Stenosis of >50% in at least one major epicardial vessel	ICA	2,234	58.5	0.685 (0.663 - 0.708)
	Fujimoto S, 2014[39]	Lesions with diameter stenosis of ≥75% were defined to be obstructive stenotic lesions. As for left main trunk lesion, lesions with diameter stenosis ≥50% were defined to be obstructive stenotic lesions.	CCTA	361	34.1	0.688 (0.626 - 0.750)
	Genders TSS, 2012[42]	Severe CAD defined as ≥70% diameter stenosis or ≥50% left main stenosis	CCTA, ICA	4,426	NA	0.78 (0.76 - 0.81)
	Jensen JM, 2012[44]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.718 (*0.674-0.762*)
	Rademaker AA, 2014[48]	>50% luminal diameter stenosis	CCTA	178	23.6	0.59 (0.51 - 0.66)
	Zhang Y, 2019[57]	≥1 lesion with ≥50% diameter stenosis	CCTA	Men: 3,001	39	0.785 (*0.767-0.803*)
				Women: 2,776	25	0.684 (*0.660-0.708*)
	Zhou J, 2017[58]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	5,743	32.6	0.772 (0.759 - 0.786)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
DFM	Genders TSS, 2011[40]	≥50% diameter stenosis in ≥1 vessel	ICA	1,683	55.7	0.78 (0.76 - 0.79)
	Jensen JM, 2012[44]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.642 (*0.596-0.688*)
	Min JK, 2015[45]	≥50% luminal diameter stenosis in any coronary artery ≥1.5 mm in diameter	CCTA	2,132	NA	0.64 (0.628 - 0.659)
	Rademaker AA, 2014[48]	>50% luminal diameter stenosis	CCTA	178	23.6	0.56 (0.49 - 0.64)
	Thomas GS 2013[51]	≥1 diameter stenosis ≥50% in a major vessel on ICA by QCA (≥1.5 mm) or CCTA (≥2.0 mm)	CCTA, ICA	431	14.6	0.69 (0.62 - 0.75)
	Voros S, 2014[53]	Outcome 50: ≥ 50% maximum diameter stenosis	CCTA	610	14	0.65 (0.59 - 0.71)
		Outcome 70: ≥ 70% maximum diameter stenosis	CCTA		NA	0.63 (0.53 - 0.73)
DFM/CASS	Adamson PD, 2018a[28]	≥70% area stenosis in any major epicardial vessel or ≥50% stenosis in the left main stem	CCTA	4,541 (PROMISE)	11.8	0.510 (0.506 - 0.514)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
			CCTA	1,619 (SCOT-HEART)	22.2	0.560 (0.548 - 0.573)
	Pickett CA, 2013[47]			1,027	6.82	0.72 (0.66 - 0.78)
	Rosenberg S, 2010[49]	≥1 atherosclerotic plaque in a major coronary artery (≥1.5 mm lumen diameter) causing ≥50% luminal diameter stenosis by QCA	ICA	526	36.5	0.663 (0.638 - 0.688)
Expanded Clinical Model	Rosenberg S, 2010[49]	≥1 atherosclerotic plaque in a major coronary artery (≥1.5 mm lumen diameter) causing ≥50% luminal diameter stenosis by QCA	ICA	526	36.5	0.732 (*0.686-0.778*)
HRA score	Yang Y, 2015[56]	High-risk CAD: left main coronary artery diameter stenosis ≥50%, 3-vessel disease (≥70%), or 2-vessel disease involving the pLAD artery	CCTA	7,333	4.8	0.71 (0.69 - 0.74)
K-score	Fujimoto S, 2014[39]	Lesions with diameter stenosis of ≥75% were defined to be obstructive stenotic lesions. As for left main trunk lesion, lesions with diameter stenosis ≥50% were defined to be obstructive stenotic lesions.	CCTA	361	34.1	0.712 (*0.656-0.770*)
Morise score	Jensen JM, 2012[44]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.681 (*0.636-0.726*)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
	Pickett CA, 2013[47]	≥50% visual luminal diameter stenosis in ≥1 epicardial coronary artery segment ≥1.5 mm in diameter	CCTA	1,027	6.82	0.68 (0.63 - 0.74)
	Rademaker AA, 2014[48]	>50% luminal diameter stenosis	CCTA	178	23.6	0.67 (0.60 - 0.74)
	Thomas GS 2013[51]	≥1 diameter stenosis ≥50% in a major vessel on ICA by QCA (≥1.5 mm) or CCTA (≥2.0 mm)	CCTA, ICA	431	14.6	0.65 (0.59 - 0.74)
PROMISE Minimal Risk model	Fordyce CB, 2017[37]	Minimal risk: normal CCTA and further conditions*	CCTA	1,528	25.0	0.713 (0.684 - 0.742)
uDFM	Adamson PD, 2018a[28]	≥70% area stenosis in any major epicardial vessel or ≥50% stenosis in the left main stem	CCTA	4,541 (PROMISE)	11.8	0.510 (0.506 - 0.514)
			CCTA	1,619 (SCOT-HEART)	22.2	0.594 (0.579 - 0.610)
	Adamson PD, 2018b[29]	Luminal cross-sectional area stenosis of ≥70% (approximating to a 50% diameter stenosis) in at least 1 major epicardial vessel or ≥50% in the left main stem.	CCTA, ICA	487	19.3	0.738 (0.687 - 0.788)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
	Almeida J, 2016[31]	Stenosis of >50% in at least one major epicardial vessel	ICA	2,234	58.5	0.664 (0.641 - 0.687)
	Baskaran L, 2018[32]	A stenosis causing ≥50% diameter stenosis	CCTA	1,738	37.7	0.767 (0.744 - 0.790)
	Bittencourt MS, 2016[33]	At least 1 segment (with a >2 mm diameter) with a lesion with ≥50% diameter stenosis	CCTA	2,274	22	0.714 (0.689 - 0.737)
	Ferreira AM, 2016[36]	Coronary diameter stenosis ≥50%	CCTA	1,069	13.8	0.70 (0.67 - 0.72)
	Genders TSS, 2011[40]	≥50% diameter stenosis in ≥1 vessel	ICA	471	NA	0.76 (0.71 - 0.81)
	Jensen JM, 2012[44]	Lumen area diameter reduction ≥50% in ≥1 coronary artery	ICA	633	34.1	0.714 (*0.670-0.758*)
	Rademaker AA, 2014[48]	>50% luminal diameter stenosis	CCTA	178	23.6	0.61 (0.53 - 0.68)
	Winther S, 2019[55]	Coronary diameter stenosis reduction ≥50% in all segments with a reference vessel diameter >2mm	CCTA	1,653	23.7	0.65 (0.61–0.68)

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Model	Study	Outcome definition	Reference Test	Sample size	Prevalence [%]	AUC (95% CI)
	Yang Y, 2015[56]	High-risk CAD: left main coronary artery diameter stenosis ≥50%, 3-vessel disease (≥70%), or 2-vessel disease involving the pLAD artery	CCTA	24,251	3.6	0.64 (0.62 - 0.67)
	Zhang Y, 2019[57]	≥1 lesion with ≥50% diameter stenosis	CCTA	Men: 3,001	39	0.782 (*0.764-0.800*)
				Women: 2,776	25	0.678 (*0.654-0.702*)
	Zhou J, 2017[58]	≥1 lesion with ≥50% diameter stenosis or any non-assessable segments due to severe calcification	CCTA	5,743	32.6	0.765 (0.751 - 0.779)
uDFM-cTn	Adamson PD, 2018b[29]	Luminal cross-sectional area stenosis of ≥70% (approximating to a 50% diameter stenosis) in at least 1 major epicardial vessel or ≥50% in the left main stem.	CCTA, ICA	487	19.3	0.757 (0.706 - 0.808)

* Further conditions are considered and should be all present, in addition to normal CCTA, for a subject to be at minimal risk: (1) coronary artery calcium score was 0 or was not obtained; (2) no evidence of atherosclerosis; (3) overall study quality was diagnostic (i.e., sufficient data quality for interpretation); (4) left ventricular function was normal or not reported; (5) no wall motion abnormalities were present or not reported; and (6) no relevant cardiovascular incidental findings that could account for the patients' symptoms (i.e., aortic dissection or pulmonary embolism) were noted. All patients with normal CCTA results were included in the minimal-risk cohort in the

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absence of any of the following adjudicated clinical events during the median 25-month follow-up period: all-cause death, nonfatal MI, unstable angina hospitalization, or revascularization during the entire follow-up period

Legend

Values in Italic are derived by the statistician (AB)

NA = Not Available

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AUC values range from 0.51[28] (almost failing) to approximately 0.81[42] (almost excellent). The statistical heterogeneity of the AUC values among the studies validating each PTP model was assessed by using the Cochran Q test and the I^2 statistic.[60] In all but two cases (CONFIRM score and Morise score) a statistically significant heterogeneity has been obtained, as expected ($p < 0.003$). On the one hand, the lack of heterogeneity is unreliable, due to the low number (≤ 5) of included studies and the low power of the Cochran Q test. On the other hand, significant heterogeneity exceeds 0.90 for most analyses and even 0.95 undermining significant interpretations ([61] and references therein). Then, in the following the discussion of the pooled values is complemented by a detailed discussion of the individual results.

From the meta-analyses, uDFM-cTn and CONFIRM show the best performances (AUC = 0.757 and pooled AUC = 0.7554 respectively). In slightly more detail, the extension of uDFM with the use of high-sensitivity cardiac troponin I (uDFM-cTn), has been validated on one only population where it showed a significantly higher AUC than uDFM alone (0.757 versus 0.738, p = 0.025) and better calibration (HL p = 0.0001 versus HL p = 0.1123).[29] The substantially steady results of the CONFIRM score on several data-sets are also confirmed on a validation data-set consisting of subjects at the low extreme of traditional cardiovascular risk factor burden.[54]

DFM, its DFM/CASS version, uDFM and Morise score show the lowest pooled AUC values <0.70. In slightly more detail, DFM/CASS has the lowest pooled AUC value (0.61) due to the two threshold-based validations reported in [28]. By excluding these values from the meta-analysis, the pooled AUC value become closer to 0.70 (0.6861, 95%CI: 0.6312-0.7409) and heterogeneity decreases to a nonsignificant level (I^2= 41.9%, p = 0.19). With regards to the DFM and its DFM/CASS version, overestimation is usually reported, especially in women.[48] However, the DFM's inferior result is also due to the fact that usually it was not carefully validated but only used as a usual reference model[44,47,48] or as a basis to establish the performances of the Corus® CAD model.[49,51,53]. The only deep validation is presented in [40]. The Morise score and the Corus® CAD are the only two models explicitly considering a female-specific factor (the oestrogen status and a sex-specific

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score, respectively): when directly compared on the same validating population, the Corus® CAD had significantly higher AUC than the Morise score (0.79 versus 0.65, $p < 0.001$).[51]

The uDFM and the CAD Consortium Clinical (CADC-clin) are the two most validated models with completely different performances (pooled AUC values: 0.6866 vs 0.7406). The uDFM updated and extended the traditional DFM to a contemporary cohort that included subjects 70 years and older. The CAD Consortium Basic model (CADC-basic) can be considered as a further update on a different contemporary population (see Table 2). The most complete validation of the uDFM, considering calibration-in-the-large, recalibration and eventually re-estimation, has been performed by the developers themselves[40] who obtained a valid overall effect of predictors. The other validating procedures limit themselves to AUC computation and to a rough assessment of under/overestimation, mainly by the Hosmer-Lemeshow goodness-of-fit (HL) test and related calibration plots (calibration-in-the-large is applied in one study[36]).

The CAD-Clin model shows good performances on validating populations by reaching estimated AUC values even > 0.80, and this high performance level is generally confirmed in other validations by taking into account estimation uncertainty (95% confidence intervals including 0.80).[32,33,50]. Moreover, its performances significantly improve with respect to the related CADC-Basic.[33,42,43,50] The pooled AUC-value (0.7406) is only slightly lower than the highest ones. It could even have been the best one if three highly performing validations[42] had presented all the data (i.e, standard error) for their inclusion in the meta-analysis. The generalizability of the CADC-Clin model to external populations was analysed by deep validation procedures [35,43,50,55] Results on miscalibration analysis could be considered quite consistent across papers. This finding indicates smaller than expected effects of the diagnostic characteristics, chest pain typicality in particular.[35,43,50] Model calibration can be worse in women compared to men, a situation that also arises from the validation of other models (e.g., DFM[40]). Despite different pooled AUC values, direct comparisons of either uDFM or CADC-Clin with the CONFIRM history-based score do not lead to a clear evaluation of the advantages of one over the other in terms of AUC,[32,36] while the

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CONFIRM score proves to be better than the DFM.[45] Figures 3 and 4 show the forest plot of the meta-analyses for uDFM and CADC-Clin model, the two most validated models. The heterogeneity for the uDFM model is not significantly reduced by removing the two threshold validations in Adamson and colleagues[28] (I^2= 95% vs I^2= 97.4%). For the uDFM and CADC-Clin models, a meta-regression analysis was also conducted which did not lead to any significant result.

The traditional DCS generally overestimates prevalence and shows a lack-of-fit by the HL test. Moreover, miscalibration results from a reduced effect of sex and chest pain typicality and an increased effect of diabetes and dyslipidaemia.[42]

The Corus® CAD model stands out from the other models because it defines an age- and sex-specific gene expression score. Validation is performed by AUC comparisons, HL test and additivity to DFM and other models. The validation procedures show significant AUC improvement when the score is added to other models (e.g., 0.81 versus 0.65 when added to Morise score, with non-overlapping confidence intervals[51]; 0.721 versus 0.663 when added to DFM, p= 0.003[49]; not shown in the table). Testing the Corus® CAD model on different data sets from an extension of the original validation population provides results very similar to the original ones.[34]

Finally, the Minimal Risk model upsets the usual point of view because it aims to directly identify patients with chest pain and normal coronary arteries. Unfortunately, the only other external validation published up to the date of our search[62] cannot be considered here because it was based on a former version of Fordyce et al. 2017[37] that included some computational errors[38].

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Discussion

External validation is an indispensable tool for investigating the generalisability of a PTP model to populations that differ from the development population study. This process can utilise different approaches, from the computation of indexes to more complex procedures that aim at understanding how the original model should adapt to the new population. The papers included in this review mainly relied on AUC. The advantage of this index lies in being suitable both for individual evaluations and for rigorous comparisons. However, the AUC is a summary: only the whole receiver operating characteristic (ROC) curve will allow evaluation of the clinical usefulness of a test by showing the true positive and false positive fractions that will be obtained for any eventually chosen cut-off.

Most of the papers included in this review did not provide a careful assessment of the discriminative performances of the validated model with respect to a well-defined threshold, but limited to compute sensitivity and specificity with respect to the thresholds suggested by either European or American guidelines. Studies on the CAD Consortium models and the Corus® CAD model are exceptions. As far as the CAD Consortium models are concerned, clinical usefulness is assessed at cut-offs that vary from 5%-20%. A cut-off of 14.75 (15 in subsequent works) was identified for the Corus® CAD model in the main work,[49] a value that corresponds to a disease likelihood of 20% on a validation data set (positivity for index $\leq$ 15). Notably, Corus® CAD recently lost Medicare coverage in the US.[63] The very low AUC values obtained by Adamson[28] at the cut-off of 15% in the comparison of the performance of major guidelines for the assessment of stable chest pain including risk-based strategies, are representative of a general clinical protection approach leading clinicians to prefer a very high sensitivity, which of course implies low specificity.[64,65]

Despite the fact that all the models are obtained by regression techniques, which allow the interpretation of the effect of the predictor on the outcome of interest, very few papers[35,40,43,50] address a complete validation procedure without rejecting a model after obtaining a poor preliminary performance on the new population by some test. Rather, a different model is developed, without any further in-depth analysis of the failure reason. Regardless of the quality of the new developed model,

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the lack of adequate consideration of in-depth validation procedures involves the loss of the information captured by the initial study and hinders a deep understanding of how effect size of relevant risk factors can change in a different geographical or setting framework.[24] For instance, deep validation procedures like miscalibration analysis allow questioning the effect of chest pain typicality in different data sets.[35,43,50] This finding is consistent with what was recently noted by Di Carli and Gupta:[66] angina remains a common presenting symptom in a high proportion of cardiac patients that do not show obstructive lesions in their coronary angiograms.

The diagnostic question is central in the determination of which diagnostic pathway and test is the most appropriate[66,67] and also affects statistical analysis. A carefully defined outcome should be required to provide a reliable basis for the evaluation of the effect of any predictive variable.[68] When referring to validation specifically, the application of a statistical model to predict an outcome different from the originally intended one raises some concerns and, eventually, should be explicitly noted. In data-driven models, the outcome definition in the population study also influences predictor selection. Thus, a small AUC value in the validation set does not necessarily indicate a lower performance of the original model on the new population. Instead, it suggests that the model may not be appropriate for the context.[62]

Despite meta-regression was not able to statistically assess the portion of heterogeneity explained by differences in sample size, prevalence and choice of the anatomical reference test, differences between studies in terms of the way the outcomes are defined and measured contribute to the methodological heterogeneity we narratively highlighted in this review.[69,70]

The main strengths of this review were the large number and high quality of included studies, the attention paid to validation procedures, as well as to AUC values alone and the careful consideration of different aspects yielding heterogeneity, as well as statistical heterogeneity alone.

The study had limitations. Most studies mainly refer to Western populations with a minority of studies referring to Asian subjects (Japan, South Corea and China).[39,45,54,56–58] Another limitation was that most of the studies did not investigate the use of any threshold. Pooled AUC values from meta-

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analyses can provide only an approximate summary of the discrimination capacity of most of the models, due to the low number of validating studies. This also affects the analysis of heterogeneity due to the low power of the test, and the feasibility of meta-regression.[71] Although the focus of our meta-analysis was not a measure of an intervention-effect, the meta-analysis was limited in the consideration of other possible sources of heterogeneity, mainly clinical like mean age or proportion of women. However, a multivariable analysis considering all the study-related variables together would have been unreliable, due to the low number of validations for most of the models.

Finally, in this review we only considered pre-test models developed in cohorts of patients referred for ICA or CCTA. Our choice was determined by main guidelines and traditional, well-established models. However, the need of models able to predict functionally significant CAD has been underlined,[72] for prognostic purposes as well. Nevertheless, how these alternative models could be used in a risk-stratification approach to guide further patients-clinician decision-making has not been assessed yet.

CONCLUSIONS

Several agencies and scientific organizations emphasise the need for increasing the knowledge on how the prediction of the disease can be modified according to the risk factors present in any specific study population or, possibly, in any particular patient. This would indeed improve the precision of the estimated clinical likelihood of CAD. However, the increasing availability of large data-sets, and the highly improved computational power seem to have directed large part of recent researches towards model development rather than model validation.[16] First of all, our review makes an important selection among the many developed models by mainly considering those externally validated. Then, it provides insights into the effects of traditional and emerging risk factors, biomarkers, and comorbidities on the PTP of obstructive CAD. Finally, our findings lead to the following important recommendations. To achieve a more robust exploitation of PTP models in decision-making processes, significant endpoints should be more clearly stated and consistently

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measured both in the derivation and validation phases. In addition, more comprehensive validation analyses should be adopted to understand model weaknesses and variations. Finally, increased efforts are still needed to thresholds validation and to analyse the effect of PTP on clinical management.

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Conflict of interests

The authors declare that there is no conflict of interest.

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Authors' contributions

AB, CGL, PM, GP and SS provided substantial contribution to the conception of the work.

CGL and PM performed the literature search and retrieved selected publications.

All the authors (PM, AB, MRT, FV, SR, GP, CC, SS and CGL) contributed to the extraction and interpretation of data.

AB carried out the meta-analysis and the meta-regression analysis.

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AB, CGL, PM and MRT assessed the quality of included studies.

All the authors contributed to draft the work. AB, CGL and PM revised it critically.

All the authors approved the version to be published and are accountable for all aspects of the work.

CGL is responsible for the overall content as guarantor.

Data sharing statement

All data used in this study are available in tables. No other data were used.

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Figure Legends

Figure 1 – Flow diagram of the study selection process.

Figure 2 – Summary of the meta-analyses. Models that were validated by one study only are denoted by AUC* and a grey color in the graphic.

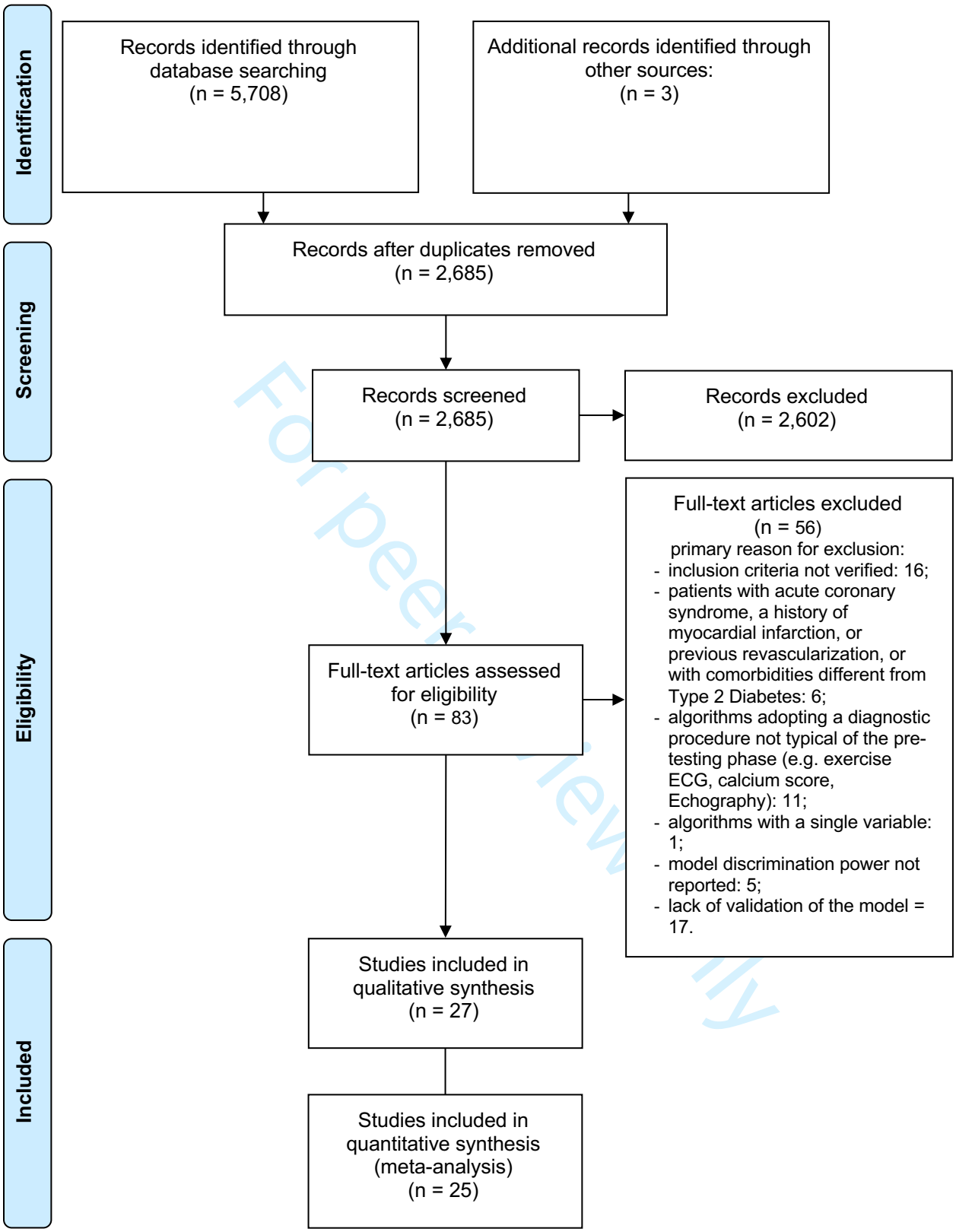
Figure 3 - Forest plot of the meta-analysis for the updated Diamond-Forrester Model.

* = PROMISE trial; ** = SCOT-HEART trial; –M = Men; -W = Women

Figure 4 - Forest plot of the meta-analysis for the CAD Consortium Clinical model.

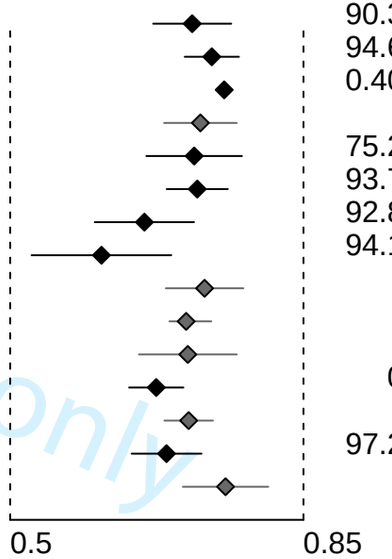
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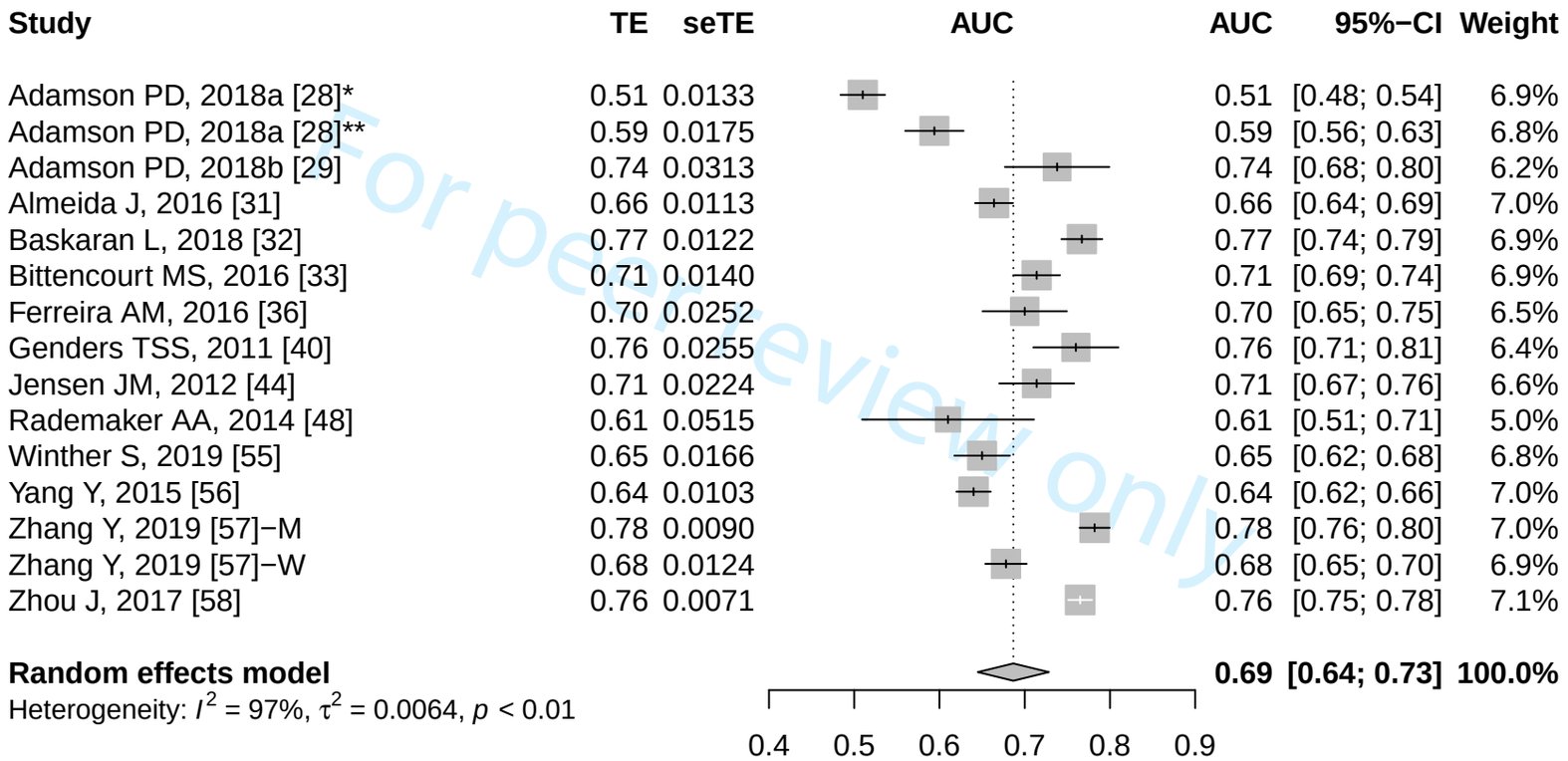


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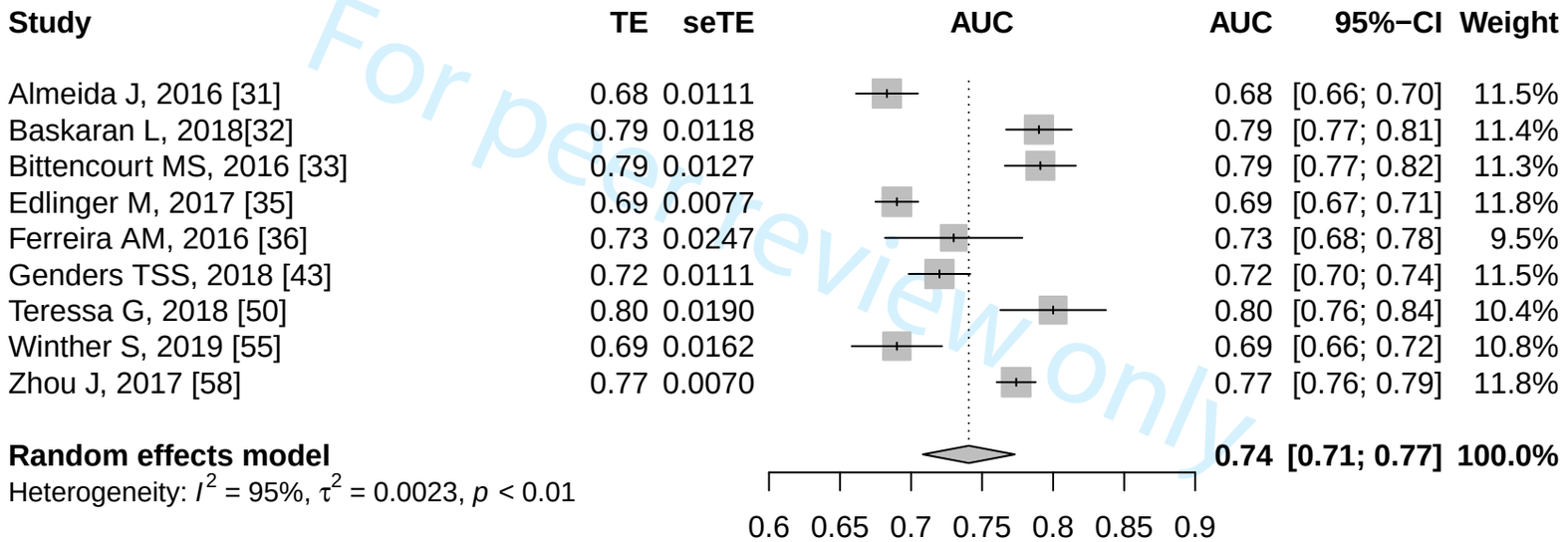
MODEL	pooled AUC	95%CI	I^2	Q−test p−value
CADC−Basic	0.7172	(0.6706, 0.7638)	90.3%	<0.0001
CADC−Clin	0.7406	(0.7083, 0.7730)	94.6%	<0.0001
CONFIRM score	0.7554	(0.7459, 0.7648)	0.40%	0.40
CORSCORE	0.727*	(0.6836, 0.7704)		
Corus® CAD	0.7196	(0.6625, 0.7766)	75.2%	0.0029
DCS	0.7233	(0.6868, 0.7597)	93.7%	<0.0001
DFM	0.6602	(0.6009, 0.7194)	92.8%	<0.0001
DFM/CASS	0.6089	(0.5255, 0.6922)	94.1%	<0.0001
Expanded Clinical model	0.732*	(0.6857,0.7783)		
HRA score	0.71*	(0.69 , 0.74)		
K score	0.712*	(0.6536, 0.7704)		
Morise score	0.6743	(0.6419, 0.7067)	0%	0.92
PROMISE minimal risk model	0.713*	(0.684, 0.742)		
uDFM	0.6866	(0.6449, 0.7283)	97.2%	<0.0001
uDFM−cTn	0.757*	(0.706 , 0.808)		



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Supplementary Material

Additional FILE 1 - Search strategy – It is the full search string adopted in OVID.

Additional File 2 – Study design and Eligibility Criteria of main studies – It provides details on the main studies cited in Table 1.

Additional File 3 – Proportion of studies with low, high or unclear risk of bias – It is a summary of the quality assessment according to QUADAS2.

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Additional FILE 1 - Search strategy

The same search strategy has been applied to the three considered databases. The number reported in each single line refers to the sum of papers retrieved in each database as these are queried at the same time.

▼ Search History (28)

# ▲	Searches	Results
1	▶ Angina Pectoris/ or Angina Pectoris.af. or Angina, Stable/ or (Stable Angina* or Chronic Angina*).af.	98847
2	▶ Chest Pain/ or Chest Pain*.af.	72609
3	▶ Coronary Heart Disease/ or (CHD or Coronary Disease* or Coronary Heart Disease*).af.	412639
4	▶ Coronary Artery Disease/ or (CAD or Coronary Artery Disease* or Coronary Arteriosclerosis or Coronary Atheroscleros*).af.	288034
5	▶ Coronary Stenosis/ or (Coronary Stenos* or Artery Stenos*).af.	64518
6	▶ or/1-5	770062
7	▶ stratification score*.af.	492
8	▶ Likelihood Functions/ or Likelihood Function*.af. or (likelihood adj5 disease).af. or CAD likelihood.af. or "predict* CAD".af.	46652
9	▶ forecasting/ or (pre-test probabilit* or PTP or predictive model* or prediction or forecast*).af.	664543
10	▶ (probability adj5 disease).af.	8725
11	▶ or/7-10	716518
12	▶ Coronary Angiography/ or Angiograph*.af.	574196
13	▶ Angiocardiography/ or Angiocardiograph*.af.	21132
14	▶ Cardiac Catheterization/ or (Cardiac Catheterization* or Heart Catheterization*).af.	122622
15	▶ Computed Tomography Angiography/ or (Coronary Computed Tomography Angiograph* or CCTA).af.	21751
16	▶ or/12-15	677008
17	▶ 6 and 11 and 16	6069
18	▶ (ANIMALS not HUMANS).sh.	7244023
19	▶ 17 not 18	5985
20	▶ limit 19 to english language or limit 19 to italian	5708
21	▶ remove duplicates from 20	2682

Additional File 2 – Study design and Eligibility Criteria of main studies

Acronym	**COMPASS**
Name	Coronary Obstruction Detection by Molecular Personalized Gene Expression (Corus CAD or ASGES)
ClinicalTrials.gov Identifier	NCT01117506
Study design	Observational prospective study. The study enrolled a patient population that presented with stable chest pain syndrome or anginal equivalent and referred for stress myocardial profusion imaging
Study Center(s)	Multicenter trial: US (7 centers)
Inclusion Criteria:	• Ages 45-90 for women; 35-90 for men. • Stable chest pain syndrome (typical or atypical) or anginal equivalent in the judgment of the investigator (e.g., pain in the neck, jaw, arm or shoulder or dyspnea possibly due to cardiac ischemia). • Referred for a stress test using Myocardial Perfusion Imaging. • The patient has signed the appropriate Institutional Review Board approved Informed Consent Form
Exclusion Criteria	• History of known MI or significant CAD. • Current MI or acute coronary syndrome. • Current New York Heart Association (NYHA) class III or IV congestive heart failure symptoms. • Severe regurgitant or stenotic cardiac valvular lesion. • Severe left ventricular systolic dysfunction (LVEF ≤ 35 % documented in the last year); if no assessment was performed or documented in the year preceding enrolment, presume normal LVEF.

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- Active systemic infection (diagnosed by a combination of clinical symptoms and laboratory testing, including but not limited to fever, leukocytosis, positive blood cultures, pneumonia, urinary tract infection, or abscess in the preceding 2 months) or chronic infection (e.g., HIV, Hepatitis B or C, Tuberculosis).
- Protocol-specified rheumatologic, autoimmune or hematologic conditions (e.g., rheumatoid arthritis, systemic lupus erythematosis, polymyalgia rheumatica, or systemic sarcoidosis).
- Known or suspected diabetes mellitus or documented Hemoglobin A1c (HbA1c) ≥ 6.5; presume normal HbA1c if none documented.
- Total WBC ≥ 11,000 cells/ul and platelet count ≤ 75,000 cells/ul from a CBC with differential drawn within 7 days prior to enrollment [WBC ≥ 11,000 cells/ul and platelet count ≤ 75,000 cells/ul from a CBC drawn > 7 days prior need to be re-drawn at enrollment].
- Recipient of any organ transplant.
- Immunosuppressive or immunomodulatory therapy including any dose of systemic corticosteroids in the preceding 2 months.
- Chemotherapy in the preceding year.
- Major surgery in the preceding 2 months.
- Blood or blood product transfusion in the preceding 2 months.
- Subjects for whom all forms (stress or pharmacologic) of MPI are contraindicated.
- Subjects for whom invasive coronary angiography or coronary CT angiography is contraindicated, including IV beta-blocker.
- Subjects who planned to decline research CCTA or invasive coronary

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	angiography, regardless of MPI result. • Subjects with history of atrial fibrillation/flutter or frequent irregular or rapid heart rhythms. • Known history of renal insufficiency (serum creatinine ≥ 2.0 mg/dL), or severe allergy to iodinated contrast.
Acronym	**CONFIRM**
Name	COroNary CT Angiography Evaluation For Clinical Outcomes: An InteRnational Multicenter Registry
ClinicalTrials.gov Identifier	NCT01443637
Study design	Observational prospective study. Patients included in the CONFIRM Registry are those that have previously undergone clinically-indicated CCTA as part of their standard of care.
Study Center(s)	Multicenter trial: North America, Europe and Asia
Inclusion Criteria:	• Age > 18 years • Evaluation by CCTA with 64-detector rows or greater for CAD evaluation as part of standard of care • Interpretable CCTA • Prospective data collection for CAD risk factors.
Exclusion Criteria	• No explicit patient exclusion criteria are defined.
Acronym	**PREDICT**
Name	Personalized Risk Evaluation and Diagnosis (Using Corus CAD or ASGES) in

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	the Coronary Tree
ClinicalTrials.gov Identifier	NCT00500617
Study design	Observational prospective study. The study enrolled patients undergoing clinically indicated invasive coronary artery angiogram or CT angiogram
Study Center(s)	Multicenter trial: US
Inclusion Criteria:	• Age 21 to 99 Years • Referral for a coronary angiogram (either invasive X-ray angiography or coronary CTA) • Any one of the following clinical syndromes: ○ chest pain syndrome, stable angina, or anginal equivalent suggesting myocardial ischemia ○ low-risk unstable angin, or ○ asymptomatic individuals with a high probability of CAD.
Exclusion Criteria	• History of myocardial infarction or known CAD • Current Myocardial Infarct (MI), acute coronary syndrome with high-risk features or unstable angina with high-risk features • New York Heart Association (NYHA) class III or IV congestive • Inability to give informed congestive heart failures • Severe left ventricular systolic dysfunction (LVEF<35%) • Severe regurgitant or stenotic cardiac valve lesion • Active or chronic systemic infection • Rheumatologic, autoimmune or hematologic conditions • Any organ transplant • Immunosuppressive therapy

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	• Chemotherapy in the preceding year • Major blood or blood product transfusion in the preceding 2 months
Acronym	**PROMISE**
Name	PROspective Multicenter Imaging Study for Evaluation of Chest Pain
ClinicalTrials.gov Identifier	NCT01174550
Study design	Interventional (Clinical Trial) randomized study
Study Center(s)	Multicenter trial: North America
Inclusion Criteria:	• New or worsening chest pain suspicious for clinically significant coronary artery disease (CAD) • no prior evaluation for this episode of symptoms • planned non-invasive testing for diagnosis • men age ≥55 years • men age ≥45 years with increased probability of coronary artery disease (CAD) due to either (A. Diabetes Mellitus (DM) requiring medical treatment OR Peripheral Arterial Disease (PAD) defined as documented >50% peripheral arterial stenosis treated medically or invasively OR cerebrovascular disease (stroke, documented > 50% carotid stenosis treated medically or invasively) OR B. At least one of the following cardiovascular risk factors: 1-Ongoing tobacco use, 2-Hypertension, 3-Abnormal ankle brachial index (ABI) defined as less than <0.9, 4-Dyslipidemia • women age ≥65 years

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	• women age ≥50 years with increased probability of coronary artery disease (CAD) due to either (A. Diabetes Mellitus (DM) requiring medical treatment OR Peripheral Arterial Disease (PAD) defined as documented >50% peripheral arterial stenosis treated medically or invasively OR cerebrovascular disease (stroke, documented > 50% carotid stenosis treated medically or invasively) OR B. At least one of the following cardiovascular risk factors: 1-Ongoing tobacco use, 2-Hypertension, 3-Abnormal ankle brachial index (ABI) defined as less than <0.9, 4-Dyslipidemia • Serum creatinine ≤ 1.5 mg/dL within the past 90 days • Negative urine/serum pregnancy test for female subjects of child-bearing potential
Exclusion Criteria	• Diagnosed or suspected acute coronary syndrome (ACS) requiring hospitalization or urgent or emergent testing; Elevated troponin or creatine kinase-myocardial band (CK-MB) • Hemodynamically or clinically unstable condition systolic blood pressure (BP) < 90 mmHg, atrial or ventricular arrhythmias, or persistent resting chest pain felt to be ischemic despite adequate therapy) • Known coronary artery disease (CAD) with prior Myocardial infarction (MI), percutaneous coronary intervention (PCI), coronary artery bypass graft (CABG) or any angiographic evidence of coronary artery disease (CAD) ≥50% lesion in a major epicardial vessel • Any invasive coronary angiography or non-invasive anatomic or functional cardiovascular (CV) test for detection of coronary artery

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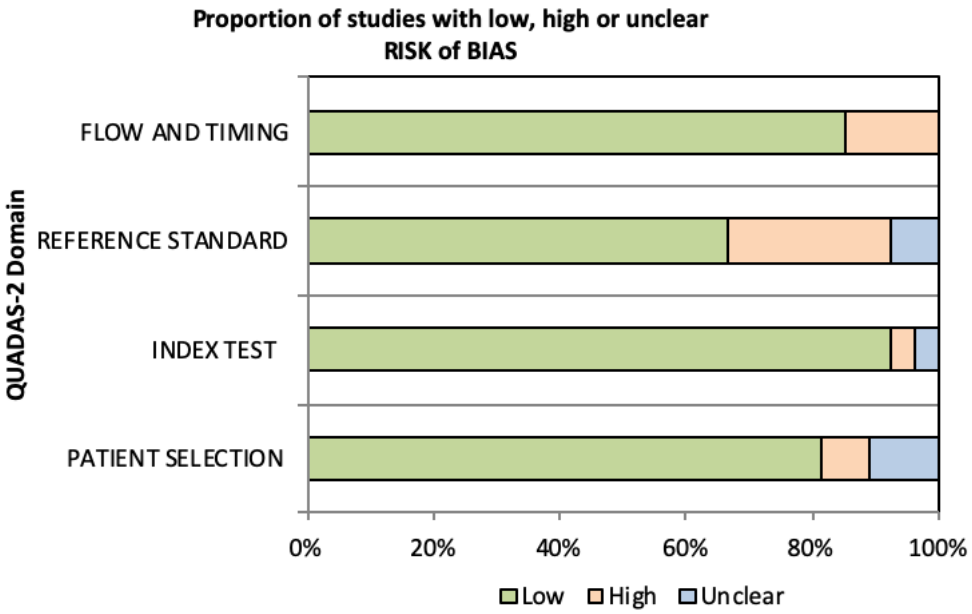
	disease (CAD), including coronary tomographic angiography (CTA) and exercise electrocardiogram (ECG), within the previous twelve (12) months • Known significant congenital, valvular (> moderate) or cardiomyopathic process (hypertrophic cardiomyopathy or reduced systolic left ventricular function (LVEF) ≤ 40%)) which could explain cardiac symptoms • Contraindication to undergoing a coronary tomographic angiography (CTA), including but not limited to: a. Allergy to iodinated contrast agent, b. Unable to receive beta blockers unless heart rate < 65 beats per minute, c. Pregnancy • Life expectancy < 2 years • Unable to provide written informed consent or participate in long-term follow-up
Acronym	**SCOT-HEART**
Name	Scottish COmputed Tomography of the HEART Trial
ClinicalTrials.gov Identifier	NCT01149590
Study design	Interventional (Clinical Trial) randomized study
Study Center(s)	Multicenter trial: Scotland (UK)
Inclusion Criteria:	• 18 and ≤75 years of age • Attendance at the Rapid Access Chest Pain Clinic
Exclusion Criteria	• Inability or unwilling to undergo computed tomography scanning, such as exceeding weight tolerance of scanner

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- Severe renal failure (serum creatinine >200 µmol/L or estimated glomerular filtration rate <30 mL/min)
- Previous recruitment to the trial
- Major allergy to iodinated contrast agent
- Unable to give informed consent
- Known pregnancy
- Acute coronary syndrome within 3 months

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Additional File 3 – Proportion of studies with low, high or unclear risk of bias



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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Additional file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7-8

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Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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